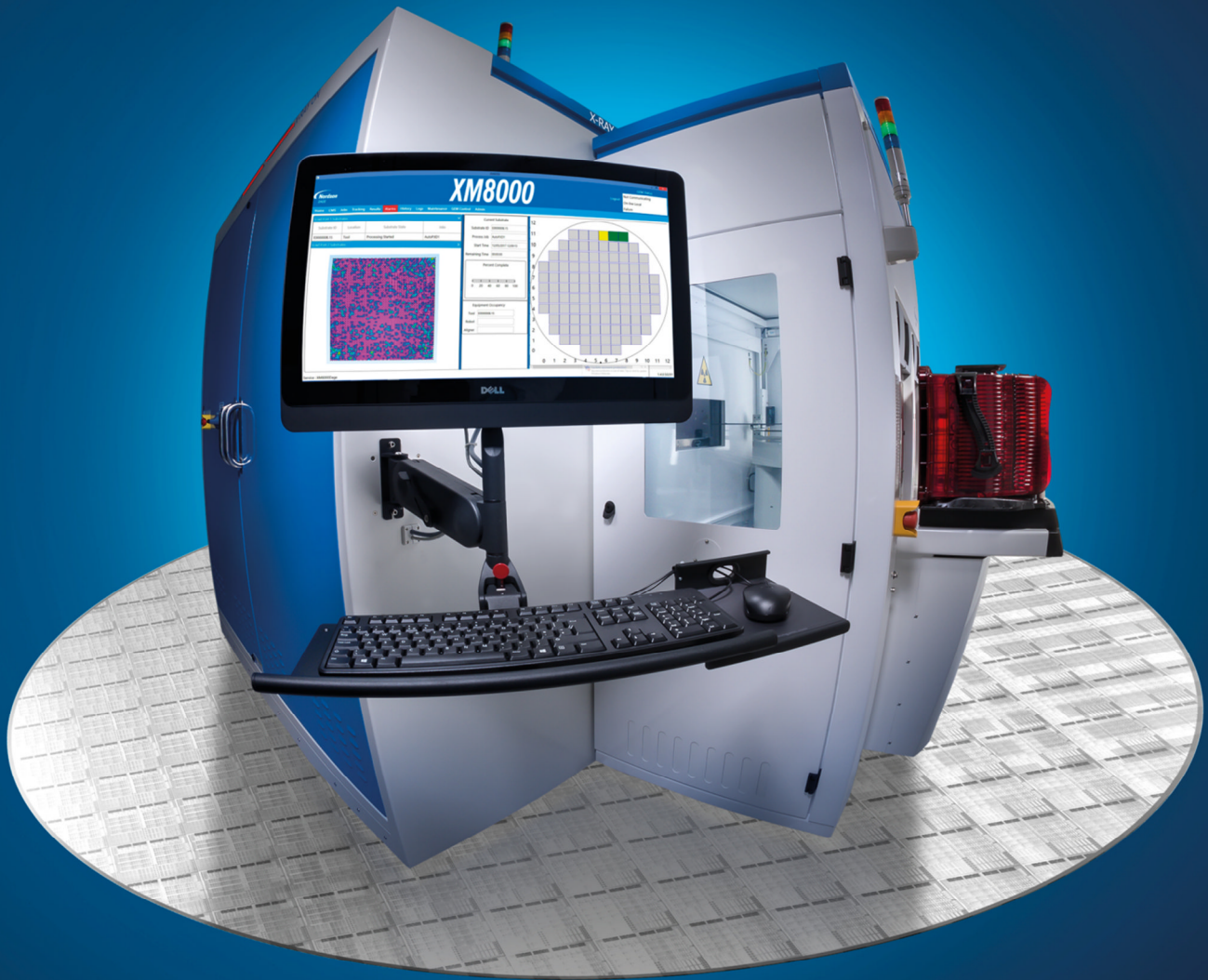


Complexity Simplified



XM8000

Intelligent X-ray Metrology

Maintenance Guide

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Nordson
TEST & INSPECTION

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Foreword

The products and equipment described in this guide are subject to continuous development and improvement. Consequently, there may be minor variations in specifications, facilities or operation that are not covered in this guide.

The installation, operation and maintenance of this equipment must only be performed by suitably trained and qualified personnel.

Every effort has been made to ensure that the information provided in this guide is accurate at the time of going to print. If any errors or omissions are noticed, please notify Dage Precision Industries Ltd.

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About this guide

This guide describes the maintenance procedures of the Nordson Test and Inspection XM8000 X-ray inspection machine.

It is intended for use by suitably-qualified and trained engineers and technical personnel.

For preinstallation and operation instruction, refer to the Preinstallation and User Guides.

Document revision: 2 EN, January 2022

Document number: DAGE-0062

The English version of this document is considered to be the original instructions.

The following documents are associated with this guide:

- XM8000 Preinstallation Guide
- XM8000 User Guide
- Nordson Test and Inspection installation overview drawing XM8000 - 803824
- Brooks JCP EFEM User Guide

- Third-party documents for:
 - PC system
 - Monitor

Design variations

Due to the Nordson Test and Inspection policy of continuous product development and improvement, some of the illustrations provided in this document may differ from the actual system you are working on. However, most of these variations are cosmetic and do not impact on the associated procedure steps. Any major differences will be mentioned, where applicable.

Radiation units

The unit of radiation measurement used throughout this guide is defined by the International System of units (SI): Sievert (Sv). However, some countries (for example, the USA), use alternative units. The conversion to REM is:

1 μ Sv per hour is equivalent to 0.1 mrem per hour.

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1. Safety information

1.1. Document signal words

The following Signal Words are used in this document. All safety related notes and reminders are marked by an appropriate symbol and the text highlighted in RED.

Please read them carefully:

Symbol	Meaning
 DANGER	DANGER indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.
 WARNING	WARNING indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.
 CAUTION	CAUTION indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices.

1.2. Labels

Safety labels are attached to the machine to indicate potential hazards.

It is **very important** you take note of these safety labels, and follow the specified safety procedures and instructions printed on them.

In certain circumstances a further label may be attached underneath, or adjacent to each safety label. This is to provide a local language translation of the safety label text.

Label	Meaning
	For the purposes of this unit, this indicates that X-rays are emitted when the X-ray tube is energized. X-rays are harmful to the human body. To comply with local and national regulations additional text may be required to accompany the radiation trefoil. Due to variations in local requirements this will need to be supplied locally.

	This indicates the presence of high voltages within the equipment. High voltages can cause death or serious injury.
--	--

	This indicates the presence of poisonous material within the machine.
---	--

Label	Meaning
 WARNING Ionising Radiation Exposure to ionising radiation when energised may cause death or serious injury. Lockout / tagout before servicing.	This indicates that X-rays are emitted when the X-ray tube is energized. X-rays are harmful to the human body.
 WARNING Toxic Lead (Pb) Lead is a poisonous metal that is harmful to the human body. Avoid direct contact with the skin.	This indicates the presence of a poisonous material within the machine. Lead is a poisonous metal that is harmful to the human body. Avoid direct contact with the skin.

Label	Meaning
 ⚠ WARNING Hazardous Voltage Enclosed Contact may cause shock, burns or death. Disconnect and lockout power before servicing.	This indicates the presence of high voltages within the equipment. High voltages can cause death or serious injury.
 ⚠ WARNING Hazardous Voltage Inside. Disconnect and lockout power before opening.	This indicates the presence of high voltages behind doors. High voltages can cause death or serious injury.
 ⚠ WARNING Live after EMO. Risk of electric shock or burn. Turn off and lockout main disconnect before servicing.	This indicates the presence of high voltages within the equipment after Emergency shut down. High voltages can cause death or serious injury.
 ⚠ WARNING Hazardous Voltage Present with Cable Disconnected. After machine power-off, residual high voltage may be present. Voltage or current sufficient to cause shock, burn or death. Before disconnecting, power down machine, lockout / tagout, then wait 5 minutes. When handling, always: Discharge metal strip to earth. Avoid contact with metal tip.	This indicates the presence of residual high voltage after power off and electrical lockout. High voltages can cause death or serious injury.
 ⚠ WARNING Moving Parts Present Can result in serious injury to hands and fingers. Keep hands away from moving parts. Lockout / tagout before servicing.	This indicates that there are moving parts / robots in this area. There is a possibility that hands or fingers may get caught, resulting in injury or crushing.
 ⚠ WARNING Mechanical Hazard Moving Parts and pinch points. Keep hands clear during operation. Lockout / tagout before servicing.	This indicates mechanical automation moving hazard. There is a possibility that hands or fingers may get caught, resulting in injury or crushing.
 ⚠ WARNING Crush Hazard Heavy object. Before withdrawing, securely fit floor support tracks. May cause death or serious injury. Approx. Weight - 750kg	This indicates that the object weighs approximately 750kg. Ensure the support tracks are securely fitted before moving.

Label	Meaning
 WARNING Heavy Object (25kg) Risk of injury or crushing. Observe all lifting precautions.	This indicates that the item weighs 25kg. Mishandling during lifting could result in injury or crushing.
 CAUTION Heavy Object Can cause muscle strain or back injury. Use lifting aids and proper lifting techniques.	This indicates that the item weighs in excess of 25kg. Mishandling during lifting could result in muscle strain or back injury.
 CAUTION Hot Surface To avoid possible skin burns allow surface to cool before touching. Lockout / tagout before servicing.	This indicates that during operation the surface may become very hot. Even after the equipment has stopped operating, some components may remain hot for some time. Confirm that they are cool before touching.

1.3. Maintenance safety rules

To avoid damage to the machine as well as avoid the risk of personal injury:

- Always** maintain the machine as prescribed in this guide
- Always** isolate the machine from electrical and pneumatic supplies using the Lock out - Tag out procedures before carrying out any maintenance work
- Always** observe all relevant local regulations and follow normal operational good practice for handling compressed air
- Always** follow the recommended cleaning procedures and clean the machine on a regular basis
- Always** use Nordson Test and Inspection approved replacement parts to maintain the machine as alternatives may result in increased radiation or degradation of the safety features

- Do not** attempt any maintenance work if you are unqualified or unfit to do so
- Do not** attempt to move or reposition the machine unless under the express guidance of an authorised and trained Nordson Test and Inspection representative
- Do not** remove the X-ray tube from the machine if you are unqualified. Unauthorized removal by untrained personnel will invalidate the warranty
- Do not** use the machine if it is damaged or faulty in any way, or where there is any doubt about its safe operation
- Do not** attempt to operate the machine outside of the specified operating parameters
- Do not** defeat any of the machine's safety features
- Do not** attempt to operate the machine with the doors open by overriding the interlocks
- Do not** compromise the integrity of the cabinet's lead shielding
- Do not** make any modifications to the machine without the express permission of Nordson Test and Inspection.

1.4. Hazardous Energies

Once the machine has been connected to the relevant service, the following hazardous energies are present within the equipment:

- Electricity
- Compressed Air
- Vacuum

And when energised:

- X-ray ionising radiation

1.5. Safety interlocks

The machine can be powered up only when:

- All Emergency Stop (EMO) pushbutton switches are in the released position.
- The machine's main electrical power isolator is in the on position.
- The key is inserted into the key-operated switch and turned to either of the 'on' positions:
 - Power Enable - the control computers, pneumatics and motors are enabled, but all X-ray systems are disabled.
 - X-Ray Enable - all machine systems are enabled, including X-ray generation. However, X-rays can only be generated when the machine is powered up, the key switch is turned to X-Ray Enable, and the interlocks on both the maintenance/service access doors and the EFEM to X-ray cabinet shutter are closed.



Note: The door and shutter interlocks are double-redundant, self-monitoring systems designed for safety critical applications.

The maintenance/service access doors are locked by a latch that can only be released by the user either via the software interface, or by operating the manual override actuator located in the electrical compartment. Either way, the lock latch releases the doors which activates the interlocks. This system:

- prevents uncontrolled opening of the doors with the X-rays on, which is extremely hazardous to health.
- turns the X-rays off.
- allows the system to bring all motion to a controlled stop.

1.6. Maintenance/service access doors

The maintenance/service access doors provide service access to the main shielded enclosure containing the servo-controlled moving axes, wafer chuck, X-ray tube and detector. During machine operation, the enclosure prevents physical access to these areas with the doors and the inner cabinet containing the necessary lead shielding to reduce the radiation leakage to less than 1uSv/hr.

Door interlocks are physically mounted on the machine enclosure and operate when the maintenance doors are opened.

You can open the maintenance/service access doors either via the software interface, or by operating the manual override actuator located in the electrical compartment, as follows:

1. Complete the electrical isolation and lockout procedure (see page 40).
2. To gain access to the electrical and pneumatic compartment, do the following:
 - Remove the upper and lower securing bolts fitted to the left-hand door.
 - Open both doors to at least 90°, and secure.
3. Locate the manual override actuator mounted centrally on the upper rear wall of the electrical and pneumatic compartment, as shown below:



4. To open the doors:
 - Depress the black pad.
 - Pull the doors open to 90°.
5. To close the doors:
 - Lift the blue plunger to reset the latch.
 - Push the doors to the fully closed position.

1.7. X-ray safety

Ionizing radiation



The machine produces X-rays which are harmful to the human body.

The X-ray source is completely contained in the shielded cabinet, with interlocking access doors preventing accidental exposure in normal use. When the interlocks are activated, or during an electrical lockout, there is no possibility of any residual ionizing radiation remaining inside the shielded cabinet. X-rays cease to be produced as soon as the power is removed from the X-ray tube.

Illuminated X-ray on warning indicators are provided on the front, side and rear panels of the cabinet.

An underlying principle of radiation protection is that the radiation from the machine should be limited to 'As Low As Reasonably Achievable'. This principle is embodied in the machine design, but this is also a requirement of the machine users, and appropriate guidance should be given in the 'safe working arrangements' or 'local rules' (see page 69).



The machine must be checked at regular intervals to verify that X-ray leakage from the shielded cabinet during operation is no higher than 1 μ Sv/H. Refer to X-ray leakage check (see page 50) for details.

If radiation is detected above this level or a fault suspected, **switch off the machine immediately.**

Shielding methods

The machine is lined with lead in all areas where X-rays could otherwise escape through the cabinet. The thickness of the lead-lining varies according to the power of the expected radiation.

The cabinet maintenance/service access doors and the EFEM to X-ray cabinet shutter are also lead-lined and overlap their respective aperture in the cabinet to ensure complete prevention of X-ray leakage. Shield bars are used extensively around the openings to reduce/eliminate radiation leakage.

X-ray On indicators

The machine is fitted with a total of five pairs of long-life LED warning indicators that are illuminated when X-rays are being generated within the cabinet. Four pairs are located around the four upper panels of the X-ray cabinet and one pair is located at the front of the EFEM unit. These indicators provide easy visibility of X-ray generation from all four sides.

1.8. Statutory regulations

The machine cabinet has been designed to comply with the appropriate US standards and UK legislation. However, the purchaser must take due account of any radiation protection legislation in their own countries. The following section reflects the requirements of UK legislation, which may or may not be appropriate in other countries.

Within the UK, please refer to 'Information to UK customers - Radiological Protection Information for X-ray Inspection System'. This document is available from Nordson Test and Inspection UK upon request.

1.9. Local safety rules

The user and any other personnel in the vicinity of the machine must observe all the information, instructions and procedures defined in the 'safe working arrangements' or 'local rules' (see page 69) in addition to the safety precautions and procedures given in this guide.

Appointed safety personnel

Safety personnel must be appointed in accordance with the safe working arrangements/local rules.

The rules may also require the appointment of an on-site Radiation Protection Supervisor (RPS) and an external Radiation Protection Advisor (RPA or Qualified Radiation Expert (QRE)).

1.10. Routine checks of X-ray cabinet

X-ray leakage from the shielded cabinet during X-ray generation must not exceed 1µSv/h.

The cabinet must be checked for radiation leakage according to the procedure prepared locally by the Radiation Protection Supervisor and agreed by the Radiation Protection Advisor.

Radiation leakage should be measured using either a Geiger Muller or CW tube suitable for the measurement of 30-160kV radiation and capable of detecting radiation at levels of less than 1µSv/h.

In the absence of any locally-prepared procedure, that detailed in X-ray leakage check (see page 50) should be performed at the stated intervals.

1.11. Hazardous material - lead

Lead is a poisonous metal that is harmful to the human body. As it may be absorbed through the skin it is important to wash hands after touching any part of the lead-lining of the cabinet. The lining is **NOT** accessible during normal use.

Lead remains in the body and the effects are cumulative.



WARNING

Lead can be harmful to the human body. Parts of the cabinet are lined with lead. Care must be taken when handling lead as it may be absorbed through the skin, or ingested by transference. After contact wash hands thoroughly using soap or detergent.

1.12. High voltage hazard

The X-ray tube installed in this machine uses high voltages to generate an X-ray beam. Residual high voltage may still be present on the HT connectors long after the machine has been switched off. Wait for at least five minutes after power off before dismantling either of the HT connectors, and always discharge the metal tip of the connector to earth before handling to ensure that no residual voltage remains.

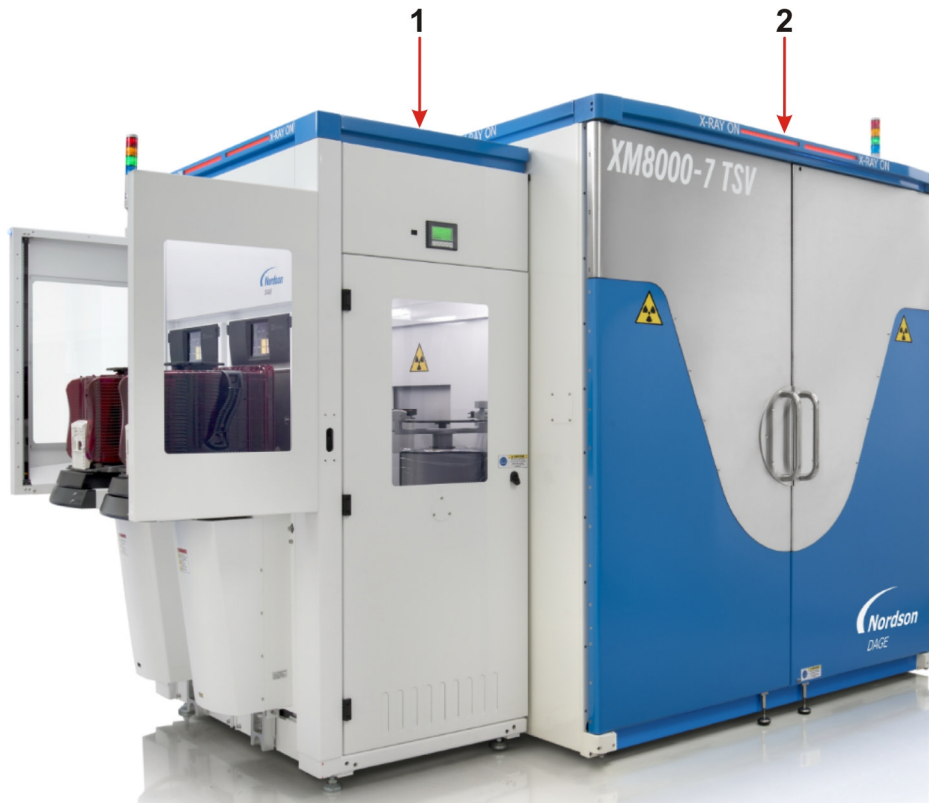
**WARNING**

High voltages can kill.

Extremely high voltages (up to 160kV) are present in this equipment. To avoid electric shocks, equipment panels or covers should be removed only by qualified personnel.

2. Machine overview

The XM8000 X-ray inspection machine consists of two main modules:



Part	Function
1 Brooks JCP EFEM wafer-handling system	This is a standard 2 bay / 2 load port Brooks Jet Common Platform (JCP) Equipment Front End Module (EFEM) configured to handle 300mm wafers. The Brooks enclosure incorporates a Class 1 Filter Fan Unit (FFU). This creates a positive pressure inside the Brooks enclosure. For full details on this system module, please refer to the separately-published Brooks JCP EFEM User Guide.
2 X-ray machine	The X-ray machine can be configured to carry out a range of manual, or fully-automatic inspection tests on the wafers delivered robotically from the EFEM. The machine enclosure is lead-lined and houses the X-ray imaging components.

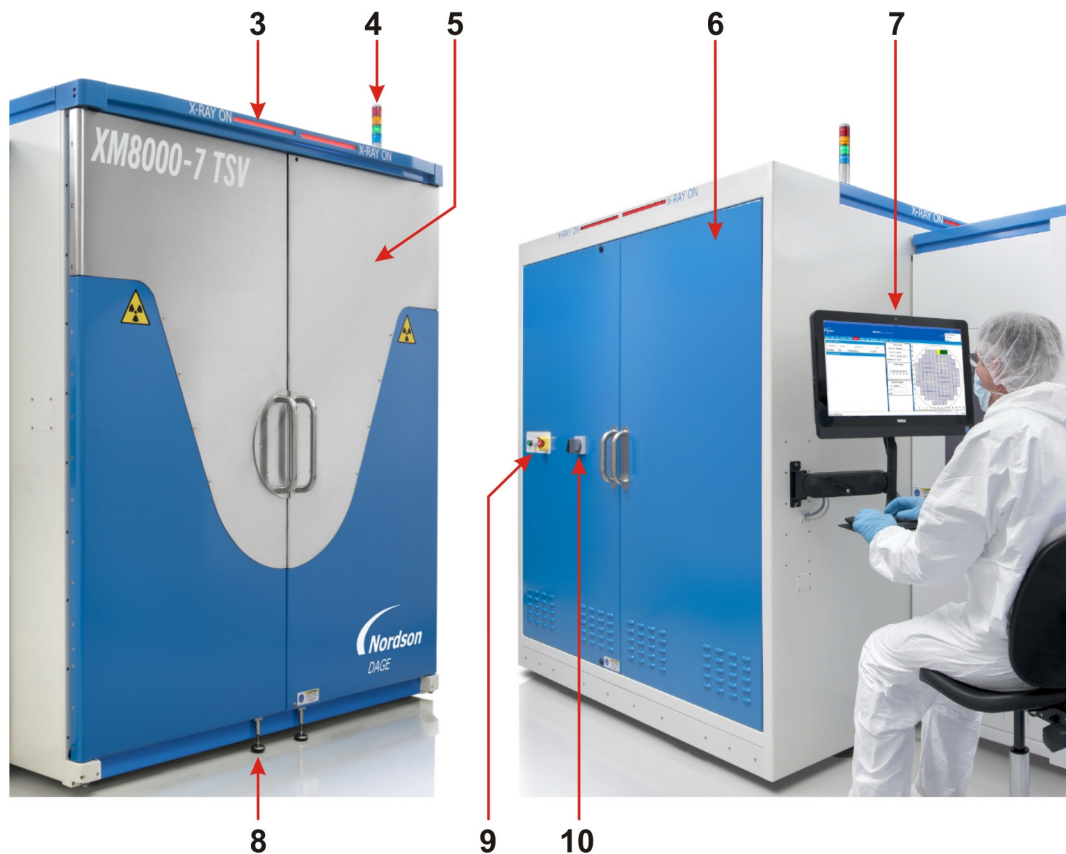
The wafers to be inspected are loaded via a Front Opening Unified/Universal Pod (FOUP).

A FOUP is a specialised plastic carrier designed to hold up to twenty-five silicon wafers securely, and to allow the wafers to be transferred safely between an Automated Material Handling System (AMHS) and the X-ray machine for processing. FOUPs typically contain a machine-readable RF tag that allows them to be identified with a unique Carrier ID.

2.1. Machine layout

The X-ray cabinet has the following main functional parts:

Note: The illustration below shows both a front and rear view of the X-ray cabinet.



Part	Function
3 X-ray warnings	The illuminated X-RAY ON warning signs on the front, side and rear walls of the machine indicate when X-rays are actually on (red).
4 Light tower	This displays the machine status using four coloured indicators, see table below for details. However, the indicators are configurable, please discuss your requirements with your local Nordson Test and Inspection representative.

Part	Function
5 Maintenance/service access doors	These doors provide full access to the interior of the lead-lined X-ray compartment. To prevent accidental opening, the doors are locked by a latch that is electrically released under software control. In the unlikely event that the door opens, the X-ray system is immediately turned off by a safety interlock system. The right-hand door is secured when closed with two front-mounted bolts.
6 Electrical and pneumatic compartment doors	The main electronic and pneumatic systems of the machine are fitted inside an accessible cabinet compartment behind hinged doors. These doors are mechanically interlocked with the main electrical isolator (10). The left-hand door is secured when closed with Allen key screws.
7 Operator workstation	This provides the means of performing and monitoring all normal operating functions. It comprises: ▪ Flat-panel monitor - connected directly to the control computer PC system, the monitor displays the operating software. ▪ Workstation tray - this supports the keyboard, and the mouse of the control computer PC system. The keyboard and mouse are used to operate all software-controlled functions of the machine. Depending on installation requirements, the operator workstation may be mounted on the EFEM cabinet.
8 Adjustable door stop	An adjustable stop is fitted on each maintenance/service access door to enable you to secure a door when open.
9 Machine control panel	This panel includes the following controls:
Power-on button	This button must be pressed in order to start the machine. It is used with the key-switch (below).

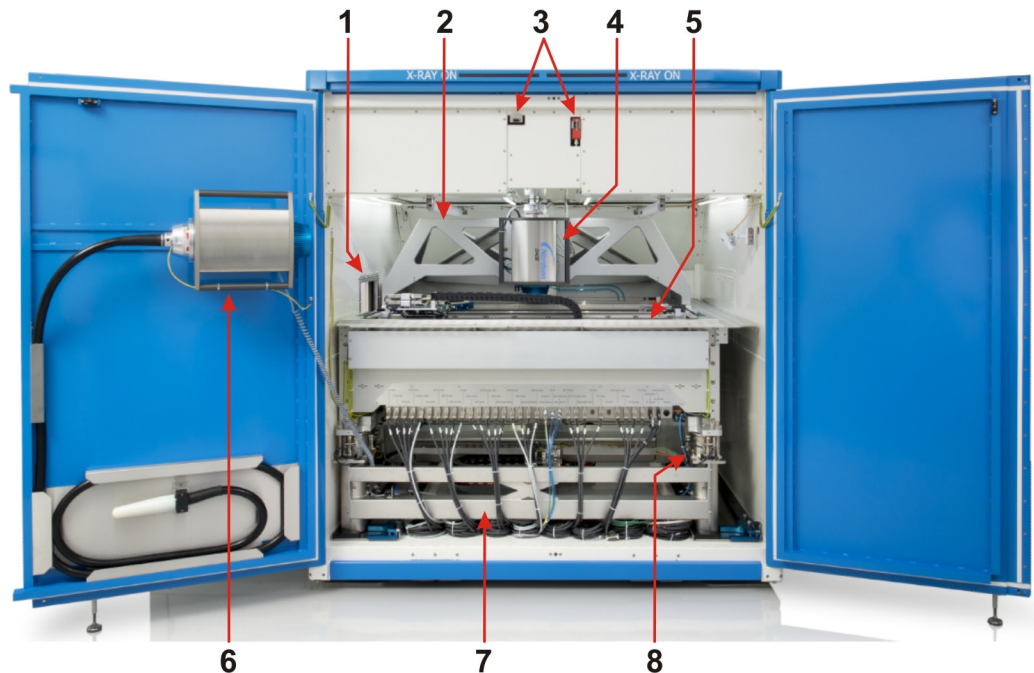
Part	Function
Key-switch	The three key-switch position functions are: Power Off - all functions of the machine are disabled. The key can be removed in this position to prevent unauthorized use. Power Enable - the control computers, pneumatics and motors are enabled, but all X-ray systems are disabled. The key can be removed in this position to prevent unauthorized use. X-Ray Enable - all functions of the machine are enabled. The key cannot be removed in this position.
Emergency Off (EMO) stop pushbutton	This emergency stop pushbutton is provided as a safety feature to be used in a hazardous situation/emergency. Operating this button instantly removes all power except the Emergency Stop pushbutton (EMO) control circuitry. Depending on machine build variant, further Emergency Off (EMO) stop pushbuttons may be mounted on the Brooks EFEM enclosure, or elsewhere on the machine cabinet. Note: To reset, rotate the red button top clockwise
10 Main electrical isolator	This lockable rotary control is used to connect or disconnect the main electrical supply to the machine.

Light tower coloured indicators

Colour	Status	Function
Red	Continuous	Currently not used.
Red	Blinking	Alarm is active.
Orange	Continuous	Load ports occupied, unavailable for transfer.
Orange	Blinking	Available for material transfer.
Green	Continuous	Idle, machine ready for use.
Green	Blinking	Job executing on machine.
Blue	Continuous	Machine is in maintenance mode.
Blue	Blinking	X-ray cabinet maintenance/service access doors open.

2.2. Interior layout

The X-ray cabinet contains the following main functional parts:

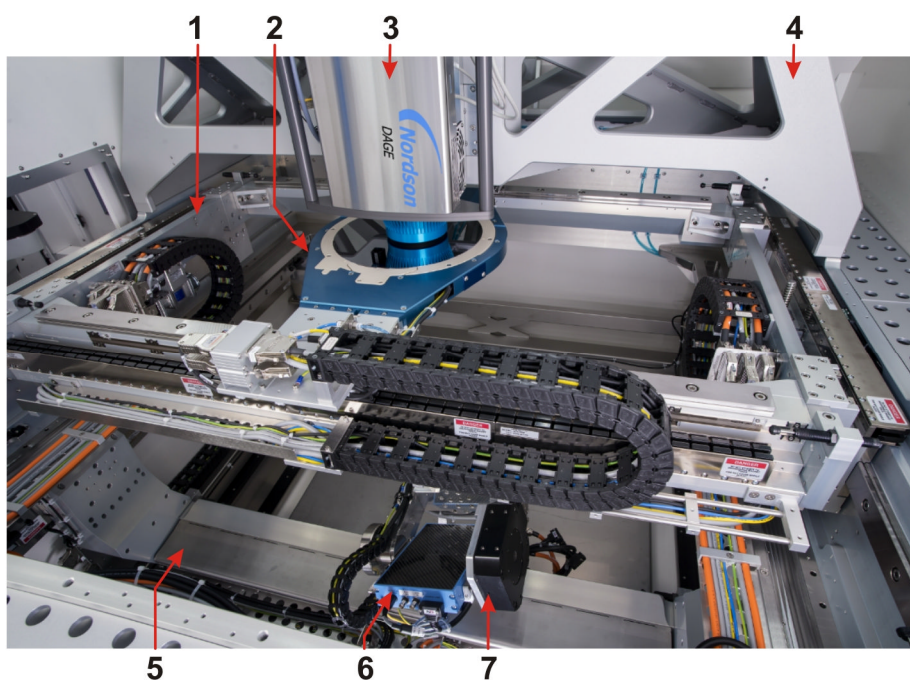


Part	Function
1 Shutter	This is a pneumatically-operated and interlocked access door between the Brooks EFEM system enclosure and the X-ray machine cabinet. When closed, it allows the user to safely use the X-ray machine while the EFEM system continues its operation cycle. When open, it allows the EFEM system to deliver a wafer to the X-ray machine wafer chuck. However, this operation can only happen if the maintenance/service access doors are closed and interlocked.
2 Upper support frame	This is a rigid aluminium frame that supports the X-ray tube and controller (4) via four anti-vibration mounts.
3 Door interlocks	There are two key-operated safety interlocks: ▪ Right-side primary - this provides an interlock with an additional solenoid that automatically locks when the maintenance/service access doors are closed. This can only be opened under software control. ▪ Left-side secondary - this provides a double-redundant interlock circuit as part of the machine safety systems.

Part	Function
4 X-ray tube and controller	The tube provides the source of X-rays and is fixed to the upper support frame (2). The tube controller is mounted on the front of the tube and contains the electronics required to control and monitor such functions as focus, voltage, and power levels.
5 Manipulator table	This assembly carries both the wafer chuck and the image detector assemblies. See below for full details on this assembly.
6 Spare X-ray tube	This is an optional spare X-ray tube.
7 Lower support frame	This is a rigid aluminium frame that supports the manipulator table (5), via four anti-vibration mounts.
8 Anti-vibration mount	These pneumatic mounts are fitted to each corner of the lower support frame and manipulator table assemblies to reduce the transmission of floor-based vibration to the X-ray imaging system components. The mounts include a height sensor/adjuster that provides for automatic self-leveling of the manipulator table under changing load conditions.

2.3. Manipulator layout

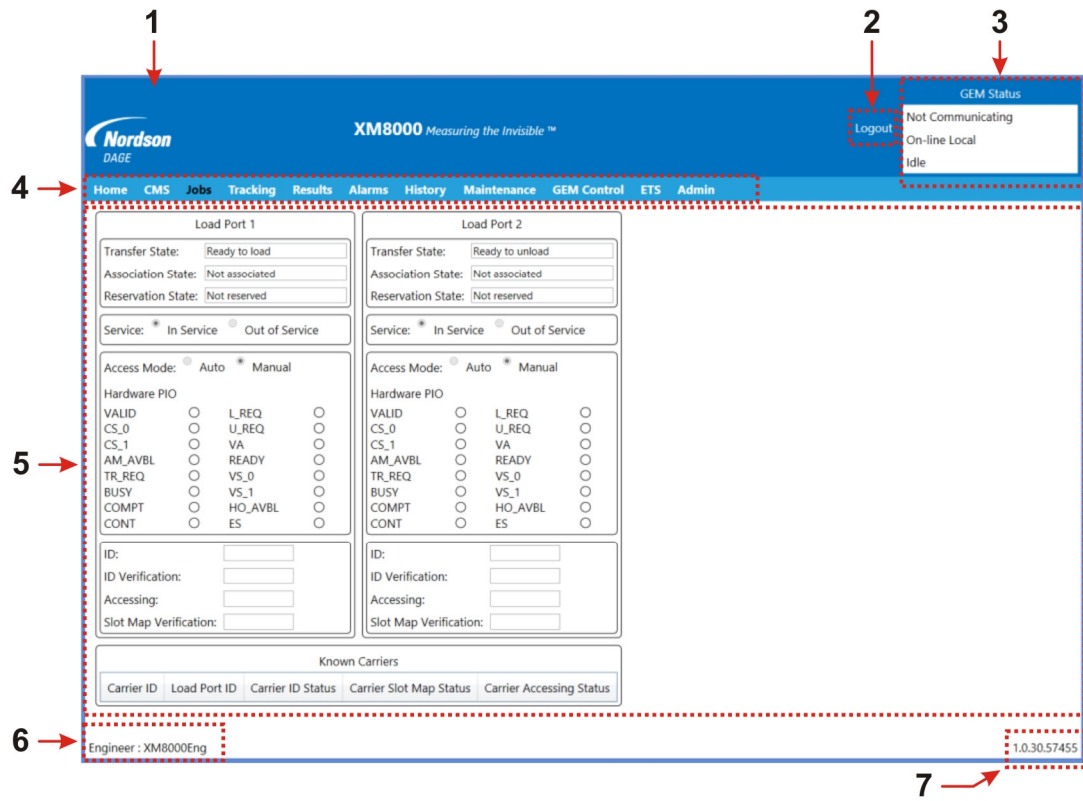
The XY manipulator mechanism contains the following main functional parts:



Part	Function
1 Wafer chuck carrier frame	This assembly carries the wafer chuck. It is mounted on precision slides and driven horizontally by linear motors in both X and Y directions. This allows different areas of the wafer sample to be accurately and repeatably positioned under the X-ray tube for examination. In addition, further rotary servo-motors are fitted to raise or lower the carrier frame Z axis.
2 Wafer chuck	This pneumatically-operated chuck receives the pre-aligned wafer delivered by the EFEM End Effector robot via the open shutter. Once delivered, the wafer is lowered under pneumatic control and clamped in place, ready for X-ray inspection.
3 X-ray tube	The tube provides the source of X-rays and is fixed to the upper support frame.
4 Upper support frame	This is a rigid aluminium frame that supports the X-ray tube and controller via four anti-vibration mounts.
5 Detector carrier frame	This assembly carries the detector carrier (7) and moves horizontally in both X and Y directions to position the image detector (6) under the wafer sample being examined. The moving assemblies of the carrier frame are mounted on precision slides and driven by linear motors.
6 Image detector	This is a CMOS high-resolution Flat Panel Detector (FPD).
7 Detector carrier	This holds the image detector (6) and is fitted to the detector carrier frame (5). Note: Depending on the system variant, the detector carrier may be optionally fitted with two motor-actuated gimbal mounts that provide both roll and pitch movement (7-axis).

2.4. Software interface

The software user-interface comprises the following functional areas:



No:	Item	Description
1	Title header bar	This identifies the application.
2	Logout control	Select this to logout the current user.
3	GEM status pane	This displays the current GEM status, such as: - Communicating - Not Communicating - On-line Local - Off-line Local - On-line Remote - Idle This pane displays a red background when the system is Not Initialized.
4	Tabbed pages bar	This displays all the available tabbed pages. Simply select the one you require and it opens in the working area. Details on the function of each tabbed page is given in the table below.

No:	Item	Description
5	Working area	This is where the software displays and operates its functional pages. The lists, controls, dialogs and other items associated with each tabbed page appear here, enabling you to carry out the necessary actions.
6	User identification	This displays the current username, and user level.
7	Software version information	This displays the software version.

Tabbed pages

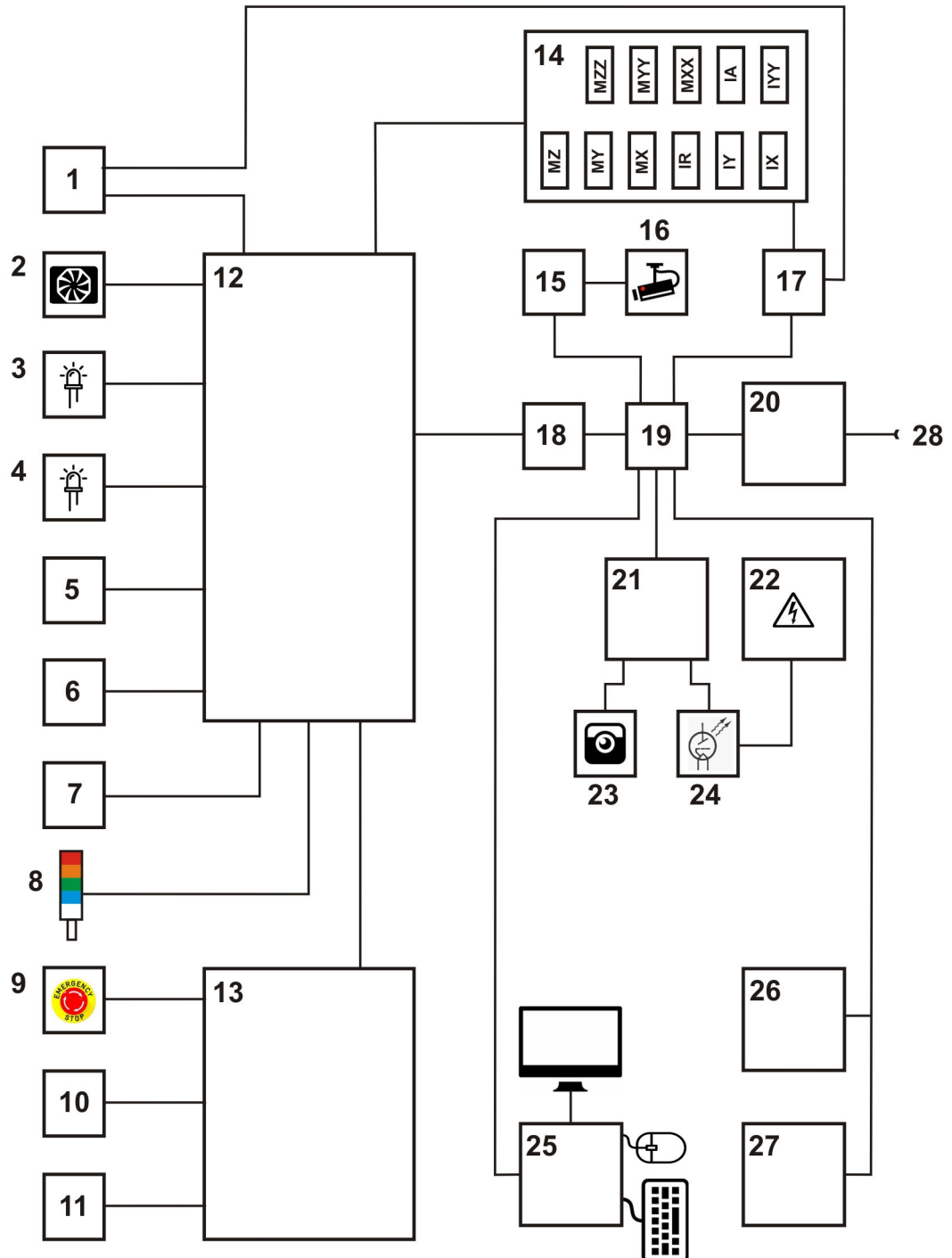
Tab	This allows you to:
Home	manually initialize and stop the five main machine components.
CMS	access the Carrier Management System page. Here you can view the status of both EFEM load ports, and any currently loaded carriers.
Jobs	Here you can: ▪ View both process and control jobs (see page 57). ▪ Define and start a manual process. ▪ Define and start an ad hoc process.
Tracking	access the Substrate Tracking page. Here you can view the current status of all material in the machine (see page 59).
Results	access the inspection results. Here you can view the .csv format measurement data files, and transfer files from the system for further analysis elsewhere.
Alarms	access the View Active Alarms page. Here you can view both active alarms and historical alarm log data.
History	access the View Run History page. Here you can view all runs, both active and historical process log data.
Maintenance	access the X-ray source Maintenance and manual control pages (see page 60). Here you can manually set various parameters and perform X-ray source maintenance procedures. Note: This page is not available to an operator-level user.

Tab	This allows you to:
GEM Control	access the SECS (SEMI Equipment Communications Standard) / GEM (Generic Equipment Model) facility host to machine communication options. Here you can enable or disable facility host communication, and configure the machine to operate either: ▪ Locally under manual control, or; ▪ Remotely under facility host control.
ETS	access the Equipment Terminal Services page. Here you can send and receive text-based messages to and from the facility host system.
Admin	access the login page. Here you can login with Operator, Engineer or Service level. It is also possible to create new users at each level and to change passwords.

3. Electrical systems

3.1. Machine electrical block diagram

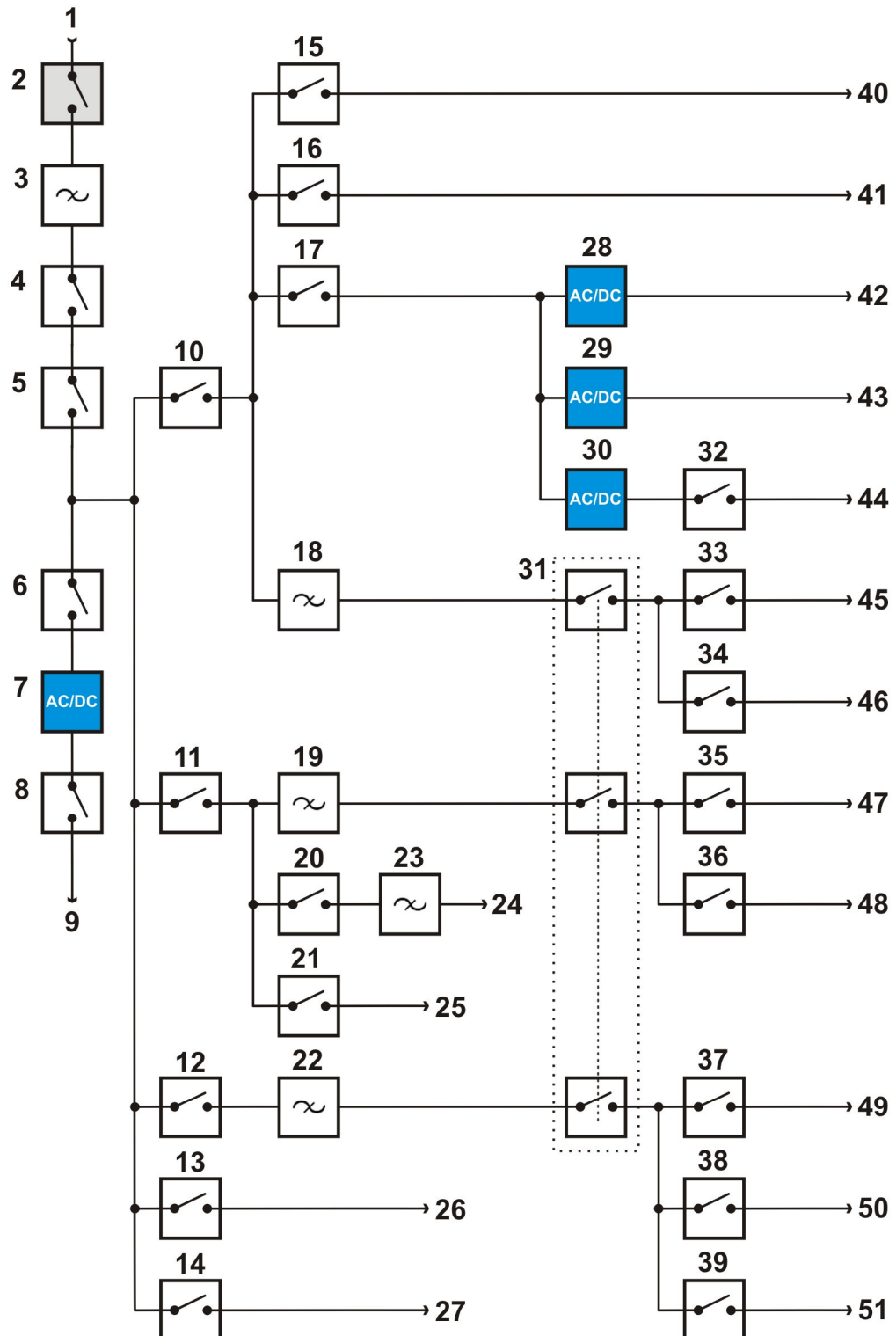
The XM8000 main electrical components and interconnections are illustrated below:



No.	Description	No.	
1	Confocal height sensor	15	Power over Ethernet (PoE) hub.
2	Filter fan unit + internal fans.	16	PoE internal camera.
3	X-ray ON indicators.	17	Ethernet hub 2.
4	Internal LED illumination.	18	OPTO 22 unit.
5	Operator panel.	19	Ethernet hub 1.
6	Wafer chuck.	20	Internal PC 2 (workflow)
7	Pneumatic control.	21	Internal PC 1 (analyser).
8	Beacons (light towers).	22	X-ray tube hi-voltage PSU.
9	EMO safety system.	23	Aspire flat panel detector.
10	Door interlocks.	24	X-ray tube.
11	EFEM to X-ray cabinet shutter.	25	Control computer (PC) + workstation.
12	Main I/O board.	26	EFEM.
13	Safety system.	27	EFEM PC.
14	Servo motor drive amplifier circuitry.	28	Facility network.

3.2. Machine electrical distribution block diagram

The functional components involved in the XM8000 electrical distribution, and their associated interconnections are illustrated below:



No.	Description	No.	
1	Mains power in.	27	External PC supply.
2	Front panel isolator switch CB1 32A.	28	Power supply 4 +12VDC (30W).
3	Main supply filter F1.	29	Power supply 3 +5VDC (25W)
4	Contactor K1.	30	Power supply 2 +24VDC (480W).
5	Contactor K2.	31	Contactors K3 + K4
6	Circuit breaker CB7 2A.	32	Circuit breaker CB21 8A.
7	Power supply 1 +24VDC (50W) EMO.	33	Circuit breaker CB11 3A.
8	Circuit breaker CB8 1A.	34	Circuit breaker CB12 2A.
9	+24VDC EMO supply.	35	Circuit breaker CB16 6A.
10	Circuit breaker CB2 8A.	36	Circuit breaker CB17 6A.
11	Circuit breaker CB3 8A.	37	Circuit breaker CB13 2A.
12	Circuit breaker CB4 6A.	38	Circuit breaker CB14 3A.
13	Circuit breaker CB5 6A.	39	Circuit breaker CB15 3A.
14	Circuit breaker CB6 6A.	40	HVPSU via contactors K5/6/7.
15	Circuit breaker CB19 4A.	41	FFU + Ioniser supply.
16	Circuit breaker CB20 6A.	42	+12VDC supply.
17	Circuit breaker CB18 3A.	43	+5VDC supply (Opto 22)
18	Filter F5.	44	+24VDC X-ray tube.
19	Filter F4.	45	MZ, MZZ servo motor feed.
20	Circuit breaker CB9 4A.	46	MY, MYY servo motor feed.
21	Circuit breaker CB10 6A.	47	IY, IYY servo motor feed.
22	Filter F3.	48	IX servo motor feed.
23	Filter F2	49	MX, MXX servo motor feed.
24	Servo amplifiers supply	50	IA servo motor feed.
25	EFEM supply.	51	IR servo motor feed.
26	Internal PC supply.		

4. Electrical safety systems descriptions

There are two elements of the machine that represent a potential danger to the operator:

- The X-ray source.
- The mechanical movements.

In normal operation there is no operator access to the areas within the cabinet that contain these hazards.

There are two primary safety systems fitted to the cabinet.

- Emergency power off (EMO) system.
- Safety interlock system.

4.1. Emergency power off

The machine is equipped with an emergency power off (EMO) system to ensure that all potential hazards are eliminated or reduced to an acceptable level when actuated, should a dangerous process or movement that may endanger equipment or people arise.

The EMO system ensures primary power is removed from the cabinet and breaks power to the servo motors and X-ray tube hi-voltage PSU when activated.

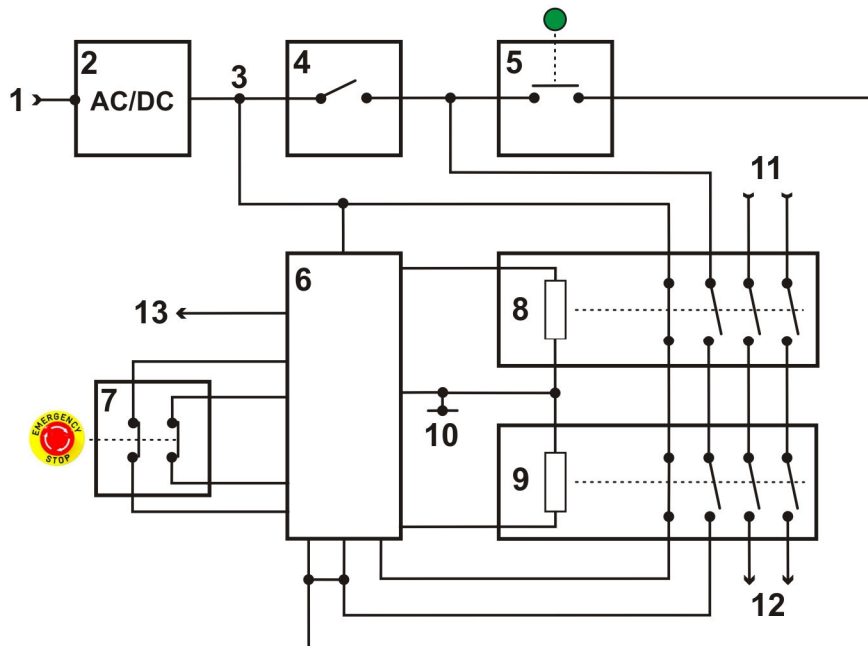
The system is designed to meet the following safety requirements:

- Application of primary power to the cabinet should not cause the machine to start.
- Operation of an emergency stop pushbutton should disconnect the primary power.
- Releasing an emergency stop pushbutton should not cause the machine to start.

If an external interface connection to the EMO system is required, **always** contact Nordson Test and Inspection for information and assistance.

Block diagram - emergency power off

The functional components of the XM8000 emergency power off (EMO) system are illustrated below:



Part	Function
1	Primary AC power feed. Via the main isolator, filter F1, and circuit breaker CB7.
2	EMO Power supply unit (PSU1).
3	EMO +24V DC feed via circuit breaker CB8.
4	Front panel key-switch.
5	Front panel start pushbutton switch.
6	EMO safety module/relay SR1.
7	Emergency stop pushbutton switch (more than one pushbutton may be fitted).
8	Contactor K1 (ganged contacts).
9	Contactor K2 (ganged contacts).
10	0V return.
11	Primary AC power feed. Via the main isolator and filter F1.
12	AC power feed to servo motor circuitry.
13	Output signal to interlock safety module/relay SR2.

EMO operation

The EMO circuit comprises:

- A dedicated EMO 24VDC PSU.
- A single-input dual-channel safety module/relay (SR1).
- Dual redundant EMO safety contactors (K1 and K2) with ganged and guided contacts.
- Machine control panel mounted emergency stop pushbutton (E-Stop) with two positive action normally-closed switches. However, depending on the machine build variant, further E-Stop pushbuttons may be fitted to the EFEM enclosure, or X-ray cabinet.

When an E-Stop is operated, the EMO system effectively removes all power from the machine, leaving only the 24VDC feed to the EMO safety circuit active.

However, the main electrical isolator, performs the same EMO function and additionally removes the 24VDC EMO PSU main AC power feed.

Power to energise the two contactors K1 and K2 is provided by the machine control panel key-switch (when set to Power Enable), and the actuation of the power-on pushbutton. Provided the safety module SR1 reports a safe condition, the two self-latching contactors are energized when the power-on pushbutton is pressed.

When contactors K1 and K2 are closed, mains power is provided to the remainder of the machine and SR1 provides an EMO monitoring function.

When an E-Stop is operated, SR1 immediately de-energises both K1 and K2 contactors, removing power (except for the EMO circuitry) from the machine.

SR1 also generates an output signal that feeds the interlock safety relay (SR2), ensuring the emergency condition is passed along the safety chain.

4.2. Safety interlock system

The safety interlock system is designed to:

- Disconnect the AC power feed to the X-ray tube hi-voltage PSU, and the +24V DC power feed to the X-ray tube electronics when either the maintenance/service access doors, or the EFEM shutter is opened.
- Disconnect the AC power feeds to the servo motors when the maintenance/service access doors are opened.
- Prevent the EFEM shutter from being opened when the maintenance/service access doors are open.
- Provide an electro-mechanical interlock signal to the EFEM Fusion controller.
- Prevent the maintenance/service access doors from opening if the interlocks have failed.

Two fully-independent interlock circuits are provided to disconnect safety critical items, so that only one circuit need operate to provide complete safety. Both circuits operate by electro-mechanical means only.

In addition, a secondary EFEM-shutter limit switch is fitted. This switch directly disconnects the AC power feed to the X-ray tube hi-voltage PSU when the shutter is opened.

4.2.1. Safety Interlock operation

The safety interlock circuit comprises:

- A door interlock dual-input dual-channel safety module/ relay (SR2).
- A primary guard-locking maintenance door interlock switch. This contains a solenoid-actuated door release.
- A secondary maintenance door interlock switch.
- Dual redundant contactors (K3 and K4) with ganged and guided contacts.
- An EFEM shutter interlock dual-input dual-channel safety module/ relay (SR3).
- A primary EFEM shutter limit switch (SW6).
- Dual redundant contactors (K6 and K7) with ganged and guided contacts.
- Contactor K5.
- A secondary EFEM shutter limit switch (SW7).

SR2 operation

If any of the safety contacts on the maintenance/service access door interlocks are opened, SR2 de-energises the contactors K3 and K4. This action:

- Removes the main AC power feeds to all servo axis motors to effectively eliminate any hazards associated with powered motorised axes.
- Signals the EFEM Fusion controller to disable all mechanical movement in the EFEM enclosure.
- Disables the shutter-up pneumatics to prevent the EFEM shutter from opening.
- Provides an output signal to SR3.

SR2 also receives an input signal from the EMO safety module/relay SR1, and consequently de-energises contactors K3 and K4 directly, when the EMO system is activated.

SR3 operation

If any of the safety contacts on the EFEM shutter limit switch are opened, SR3 de-energises the contactors K6 and K7. This action removes the main AC power feed to the X-ray tube hi-voltage PSU, and the +24V DC power feed to the X-ray tube electronics. This immediately halts and disables the generation of X-rays.

SR3 also receives an input signal from the safety module/relay SR2, and consequently de-energises contactors K6 and K7 directly if either of the following occur:

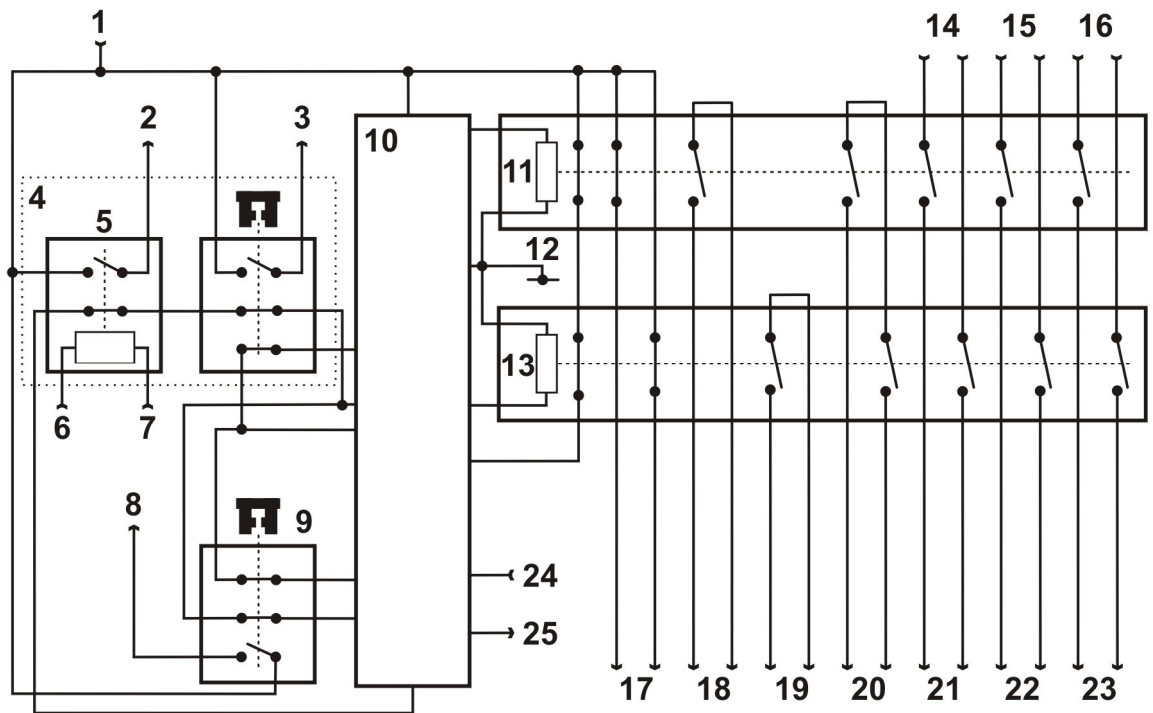
- The EMO system is activated.
- Any of the maintenance/service access door interlock switch safety contacts are opened.

Contactor K5 operation

The single contactor K5 provides a monitor signal that indicates the state of the AC feed supplied via contactors K6, K7, and the EFEM shutter limit switch to the X-ray tube hi-voltage PSU. The physical operation of contactors K5, K6 and K7 is checked by the safety module/relay SR3.

4.2.2. Block diagram - door interlock wiring

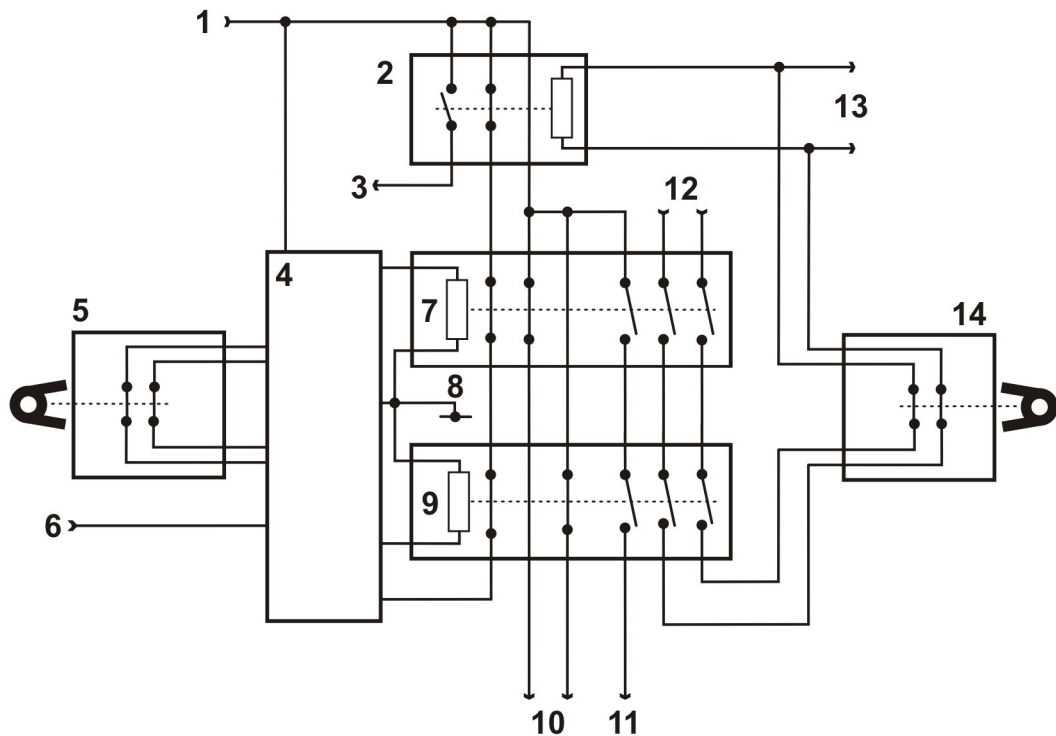
The functional components of the XM8000 door interlock system are illustrated below:



No.	Description	No.	Description
1	+24VDC supply in from PSU2 via circuit breaker CB21	14	AC power via circuit breaker CB2
2	Door solenoid monitor signal.	15	AC power via circuit breaker CB4
3	Door interlock monitor signal.	16	AC power via circuit breaker CB3
4	Right hand door interlock switch.	17	Contactor monitoring signals
5	Door locking solenoid.	18	EFEM Fusion + interlock in/out
6	Door locking solenoid +ve feed.	19	EFEM Fusion - interlock in/out
7	Door locking solenoid -ve feed.	20	Shutter up signal feed and return
8	Door interlock monitor signal.	21	MZZ, MZ, MYY, MY servo power feeds.
9	Left hand door interlock switch.	22	MXX, MX, IA, IR servo power feeds.
10	Safety interlock module/relay SR2.	23	IYY, IY, IX servo power feeds.
11	Contactor K3.	24	Input signal from EMO module/relay SR1.
12	0V return.	25	Output signal to SR3.
13	Contactor K4.		

4.2.3. Block diagram - EFEM shutter interlock wiring

The functional components of the XM8000 shutter EFEM interlock system are illustrated below:



No.	Description	No.	
1	+24VDC supply in from PSU2 via circuit breaker CB21	8	0V return.
2	Contactors K5	9	Contactors K7
3	Contactors monitor signal	10	Contactors monitor signals
4	Safety interlock module/relay SR3.	11	X-ray tube electronics +24V supply
5	EFEM shutter limit switch SW7	12	AC power via circuit breaker CB19
6	Input signal from safety module/relay SR2.	13	X-ray tube hi-voltage PSU AC supply
7	Contactors K6	14	EFEM shutter limit switch SW6

4.2.4. Contactor failure

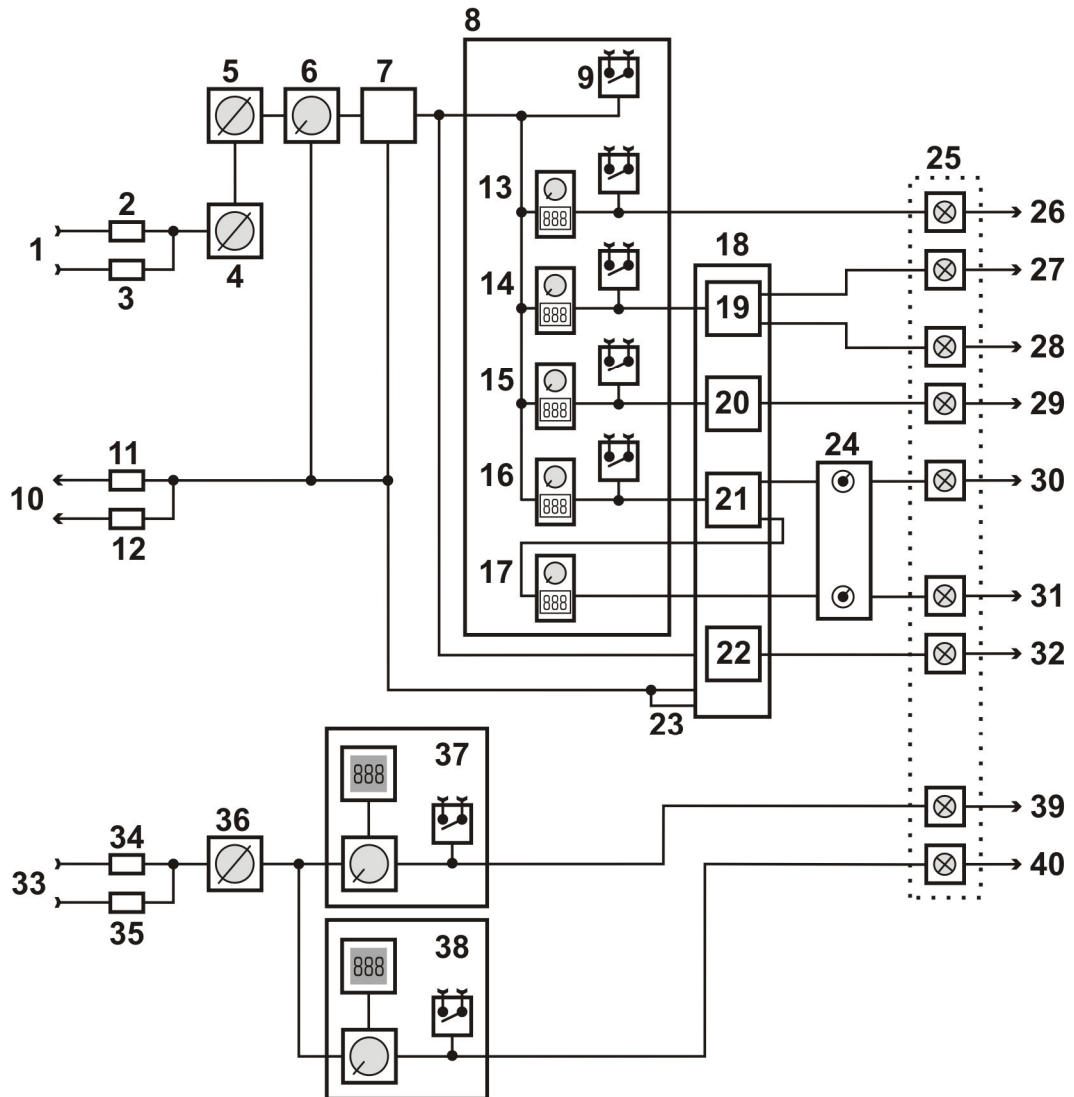
The contactors described in this section are specifically designed to operate with very high levels of integrity. There are two possible failure scenarios:

- Contact fails to close - This is intrinsically safe as the interlocks depend on contact closure, and therefore would alert the operator to the fault condition.
- Contact fails to open - Contacts from K3 and K4, and K6 and K7, are connected in series - therefore should one in-circuit contactor fail, the safety function is performed by the second. The failure of a contactor is detected by the safety interlock relay module preventing continued operation. A single failure is detected by the safety system.

5. Pneumatic systems

5.1. Pneumatic system block diagram

The XM8000 main pneumatic components and interconnections are illustrated below:



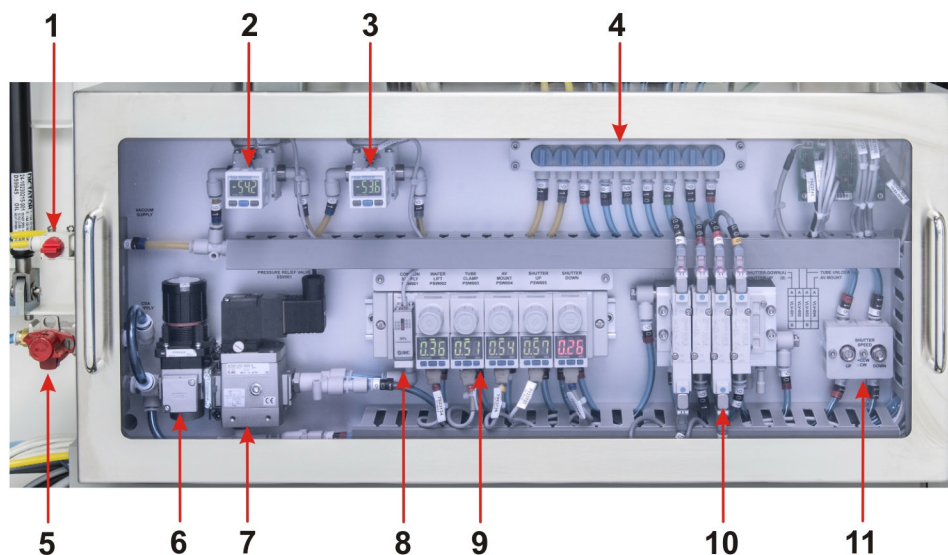
No.	Description	No.	Description
1	Main compressed air supply in (factory)	14	Tube clamp air regulator, digital level gauge + switch PSW003
2	Air supply push-fit connector - roof	15	AV mount air regulator, digital level gauge + switch PSW004
3	Air supply push-fit connector - base	16	Shutter up supply air regulator, digital level gauge + switch PSW005
4	Key-lockable externally-mounted air supply isolator valve	17	Shutter down air regulator and digital level gauge
5	Internally-mounted air supply lockable isolator valve	18	Valve manifold

No.	Description	No.	Description
6	Air supply pressure regulator	19	Tube clamp valve - VLV003
7	Air supply soft start pressure relief valve - SSV001	20	AV mount valve - VLV004
8	Air distribution manifold	21	Shutter up valve - VLV001
9	Air supply common pressure switch - PSW001	22	Shutter down valve - VLV002
10	Air supply exhaust	23	Valve manifold exhaust
11	Air supply exhaust push-fit connector - roof	24	Dual flow rate controls (shutter up/down)
12	Air supply exhaust push-fit connector - base	25	Independent air/vac isolator valves
13	Wafer lift air regulator, digital level gauge + switch PSW002	26	Wafer lift air supply

No.	Description	No.	Description
27	Tube clamp air supply 1	34	Vac supply push-fit connector - roof
28	Tube clamp air supply 2	35	Vac supply push-fit connector - base
29	AV mount air supply	36	Vac supply isolator valve
30	EFEM shutter up air supply	37	EFEM vacuum regulator, digital level gauge and switch
31	EFEM shutter down air supply	38	Wafer chuck vacuum regulator, digital level gauge and switch
32	EFEM shutter exhaust	39	EFEM vacuum supply
33	Main vacuum supply in (factory)	40	Wafer chuck vacuum supply

5.2. Pneumatic system components

The main XM8000 pneumatic components are mounted on a chassis located within the electrical and pneumatic compartment.

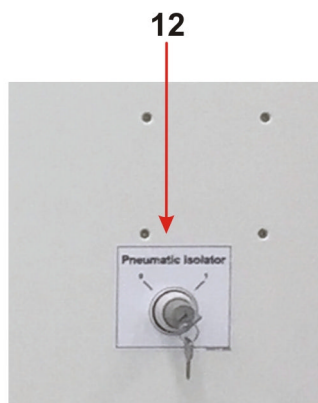


No.	Component/Function
1	Vacuum isolator valve (factory supply)
2	EFEM Vacuum regulator, digital level gauge and switch - VSW001
3	Wafer chuck vacuum regulator, digital level gauge and switch - VSW002
4	Independent isolator valves for: ▪ Wafer chuck lift ▪ Tube clamp ▪ AV mounts ▪ Shutter up/down

No.	Component/Function
5	Lockable compressed air isolator valve. This is fed directly from the externally-mounted key-lockable main factory supply isolator (see below)
6	Air supply pressure regulator
7	Air supply soft start pressure relief valve - SSV001
8	Air supply common pressure switch - PSW001
9	Individual air regulators with digital level gauges for: ▪ Wafer lift + switch PSW002 ▪ Tube clamp + switch PSW003 ▪ AV Mount + switch PSW004 ▪ Shutter up + switch PSW005 ▪ Shutter down
10	Valve block, comprising: ▪ Shutter up valve - VLV001 ▪ Shutter down valve - VLV002 ▪ Tube clamp valve - VLV003 ▪ AV mount valve - VLV004
11	EFEM shutter up/down air flow rate controls

Main CDA Isolator

The main key-lockable factory supply compressed air (CDA) isolator valve (12) is typically fitted to the XM8000 X-ray cabinet adjacent to the EFEM.



Tube colour coding

Within the machine enclosure, all pneumatic tubing is colour-coded, as follows:

- Vacuum - yellow
- Compressed air (CDA and exhaust) - blue

6. LOTO procedures



WARNING

Always complete the following procedures to isolate the electrical and pneumatic supplies if maintenance work is to be carried out on the machine, or the machine needs to be made safe for any other reason.



WARNING

Nordson Test and Inspection advise all customers and users that there are NO user-serviceable components within the XM8000 cabinet enclosure. Therefore, there is generally no need to open either the maintenance/service access or electrical and pneumatic compartment doors at any time, for any reason.

ALL maintenance work that involves the opening of doors must be carried out by Nordson Test and Inspection engineers only.



- Electrical isolation and lockout (see page 40)



- Pneumatic isolation and lockout (see page 43)

Once the maintenance work has been successfully completed and/or the machine needs to be returned to service, the following procedures should be carried out to re-establish the electrical and pneumatic supplies:

- Re-establishing the pneumatic supply (see page 45)
- Re-establishing the electrical supply (see page 46)

6.1. Electrical isolation and lockout



Main incoming electrical supply to the main isolator is always live, unless the main facility supply to the machine is disconnected and isolated.

Note: Before proceeding with this procedure, always ensure that every member of the LOTO group has a fully working padlock and corresponding keys.

Important: Always test the padlocks and keys BEFORE fitting to any LOTO devices!

Procedure requirements:

Personnel	Minimum 2. One person should be elected as the group lead.
Time	30 minutes.
Tools	▪ Metric Allen keys. ▪ Cleanroom pen.
Parts	▪ Group LOTO device, padlock and tag. ▪ Group LOTO lockbox. ▪ Personal LOTO padlocks and tags. (See NOTICE above)
PPE	As per site requirements for normal working.
Prerequisite skills	▪ XM8000 engineer-level training ▪ Site safety training. ▪ LOTO training.

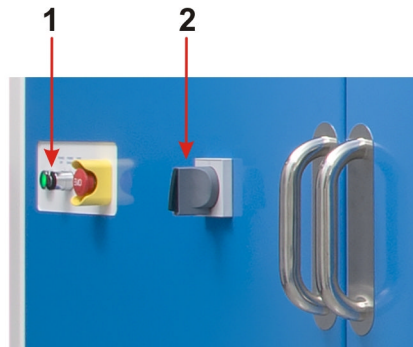
Preparing for electrical isolation and lockout

If not already completed, do the following:

1. Ensure local approval and permission has been authorized and obtained.
2. Ensure that the machine is empty of wafers, and idle.
3. At the operator workstation, login at Engineer level.
4. Select the Maintenance tab.
5. Select Enter Maintenance.
6. Select the Recovery page.
7. Select Stop Components.
8. Select Shutdown PCs.
9. Wait three minutes (minimum) for all system PCs to shutdown.

10. At the electrical and pneumatic compartment doors:

- Turn the key-operated switch (1) on the machine control panel to Power Off.
- Switch off the machine's power isolator (2) by rotating counter-clockwise.



11. Isolate the facility's electrical supply to the machine by switching off the appropriate power isolator, or circuit breaker, wherever possible.
12. Cordon off the area as per any site safety requirements.

Performing the lockout

1. The LOTO group lead to fill out the appropriate group tag (or tags) in BLOCK CAPITALS, taking care to include all the required details, such as: full name and date (as a minimum). However, please note that your LOTO group tag may include other required details.
2. The LOTO group lead to:
 - Confirm that the facility's electrical supply isolator is fully in the 0 or OFF position where applicable, or the facility's electrical supply circuit breaker is switched off.
 - Confirm that the machine's electrical isolator is fully in the 0 or OFF position.
 - Fit the appropriate group LOTO device and group tag to the isolator(s).
 - Make sure that the tag information is clearly visible.
 - Fit and lock the group padlock through the group device. See the example illustration below:



3. Each member of the LOTO group to fit and lock his or her personal padlock through the group device.

4. The LOTO group lead to do the following:
 - Confirm that the facility's electrical isolator is fully locked in the 0 or OFF position, where applicable.
 - Confirm that the machine's electrical isolator is fully locked in the 0 or OFF position.
 - Place both sets of keys for the group padlock into a lockbox.
 - Place keys for all LOTO group member padlocks into the lockbox.
 - Attach a personal tag and padlock on the lockbox, taking care to identify themselves as the LOTO group lead, with his or her full name, contact details and date.
5. If you need to gain access to the electrical and pneumatic compartment, do the following:
 - Remove the upper and lower securing bolts fitted to the left-hand door.
 - Open both doors to at least 90°, and secure.

Important: The machine's electrical isolator is a two-part device. When the door carrying the actuating handle is opened, the isolator unit separates. The main body of the isolator is fitted in the electrical compartment and is actuated by a shaft. Therefore, rotating this shaft allows mains power to be applied to the equipment - if the supply is not completely isolated at the facility.

6.2. Pneumatic isolation and lockout

Note: Before proceeding with this procedure, always ensure that every member of the LOTO group has a fully working padlock and corresponding keys.

Important: Always test the padlocks and keys BEFORE fitting to any LOTO devices!

Procedure requirements:

Personnel	Minimum 2. One person should be elected as the group lead.
Time	30 minutes.
Tools	▪ Metric Allen keys. ▪ Cleanroom pen.
Parts	▪ Group LOTO device, padlock and tag. ▪ Group LOTO lockbox. ▪ Personal LOTO padlocks and tags. (See NOTICE above)
PPE	As per site requirements for normal working.
Prerequisite skills	▪ XM8000 engineer-level training ▪ Site safety training. ▪ LOTO training.

Preparing for pneumatic isolation and lockout

1. Ensure local approval and permission has been authorized and obtained.
2. Complete the electrical isolation and lockout procedure (see page 40).
3. Isolate the facility's compressed air supply to the machine by rotating the main isolation valve counter-clockwise to the closed (0) position.
The facility (CDA) isolator valve is typically fitted to the XM8000 X-ray cabinet adjacent to the EFEM.
4. Remove the key from the facility CDA isolation valve, and retain safely.
5. To gain access to the electrical and pneumatic compartment, do the following:
 - Remove the upper and lower securing bolts fitted to the left-hand door.
 - Open both doors to at least 90°, and secure.

6. Locate the air isolator valve fitted to the left-hand side of the pneumatic chassis and rotate the red-coloured knob counter-clockwise to close the valve and depressurise the downstream circuit.



7. Cordon off the area as per any site safety requirements.

Performing the lockout

1. The LOTO group lead to fill out the appropriate group tag in BLOCK CAPITALS, taking care to include all the required details, such as: full name and date (as a minimum). However, please note that your LOTO group tag may include other required details.
2. The LOTO group lead to:
 - Confirm that the facility CDA isolation valve is in the closed (0) position, and that the key has been removed.
 - Confirm that the pneumatic chassis air isolator valve is fully in the CLOSED position.
 - Fit the appropriate group LOTO device and group tag to the isolator valve.
 - Make sure that the tag information is clearly visible.
 - Fit and lock the group padlock through the group device. See the example illustration below:



3. Each member of the LOTO group to fit and lock his or her personal padlock through the group device.
4. The LOTO group lead to do the following:
 - Place both sets of keys for the group padlock into a lockbox.
 - Place keys for all LOTO group member padlocks into the lockbox.
 - Place the facility CDA isolation valve key into the lockbox.
 - Attach a personal tag and padlock on the lockbox, taking care to identify themselves as the LOTO group lead, with his or her full name, contact details and date.

6.3. Re-establishing pneumatic supply



To prevent unintended restoration of the supply, this procedure should be carried out under the exclusive supervision of the person carrying out the work on the machine, and the group LOTO lead.

Procedure requirements:

Personnel	Minimum 2. One person should be elected as the group lead.
Time	15 minutes.
Tools	Metric Allen keys.
Parts	- Group LOTO device, padlock and tag. - Personal LOTO padlocks and tags.
PPE	As per site requirements for normal working.
Prerequisite skills	- XM8000 engineer-level training - Site safety training. - LOTO training.

- The LOTO group lead to:
 - Remove his or her personal padlock and tag from the lockbox.
 - Retrieve the LOTO group padlock keys, and distribute to respective group members.
 - Retrieve the facility CDA isolation valve key.
- Each member of the LOTO group to unlock and remove his or her padlock from the group device.
- The LOTO group lead to:
 - Unlock and remove the group padlock from the group LOTO device.
 - Remove the group LOTO device and group tag from the pneumatic chassis air isolator valve.
- Rotate the red-coloured knob of the pneumatic chassis air isolator valve clockwise to the OPEN position.
- Close the electrical and pneumatic compartment doors.
- Refit and tighten the upper and lower securing bolts to the left-hand door.
- Refit the key to the facility CDA isolation valve and rotate clockwise to the open (1) position.
- Complete the Re-establishing electrical supply procedure (see page 46).

6.4. Re-establishing electrical supply



To prevent unintended restoration of the supply, this procedure should be carried out under the exclusive supervision of the person carrying out the work on the machine, and the group LOTO lead.

Procedure requirements:

Personnel	Minimum 2. One person should be elected as the group lead.
Time	15 minutes.
Tools	Metric Allen keys.
Parts	▪ Group LOTO device, padlock and tag. ▪ Personal LOTO padlocks and tags.
PPE	As per site requirements for normal working.
Prerequisite skills	▪ XM8000 engineer-level training ▪ Site safety training. ▪ LOTO training.

1. The LOTO group lead to:
 - Remove his or her personal padlock and tag from the lockbox.
 - Retrieve the LOTO group padlock keys, and distribute to respective group members.
2. Each member of the LOTO group to unlock and remove his or her padlock from the group device(s).
3. The LOTO group lead to:
 - Unlock and remove the group padlock from the group LOTO device(s).
 - Remove the group LOTO device and group tag from the facility's electrical supply isolator, if applicable.
 - Remove the group LOTO device and group tag from the machine's electrical isolator.
 - Re-establish the facility's electrical supply to the machine by switching on the appropriate power isolator, or circuit breaker, if applicable.
4. If not already done:
 - Close the electrical and pneumatic compartment doors.
 - Refit and tighten the upper and lower securing bolts to the left-hand door.
5. Switch on the machine's electrical isolator by rotating clockwise.

-
6. At the machine control panel:
 - Turn the key-operated switch to Power Enable.
 - Push the green power-on button.

The machine will restart.
 7. For continued X-ray protection, perform an X-ray leakage test if **ANY WORK WHATSOEVER** has been undertaken on the machine.
 8. Verify that the machine is restored back to normal operation and in particular the functioning of any part serviced or replaced.
 9. Inform all affected parties about lockout tag removal and that the machine is restored to full use.

7. Routine preventative maintenance

Note: The XM8000 contains NO user-serviceable/replaceable components. Please contact your Nordson Test and Inspection representative for assistance.

Maintenance schedule

Nordson Test and Inspection recommend that the following preventative maintenance schedule is adopted to ensure continued trouble-free operation. Any issues identified during the various inspection checks or tests should be investigated prior to continued use of the machine by a suitably-trained competent person.

The intervals specified here should be regarded as a minimum. However, more frequent intervals may be recommended by your local and/or company regulations, particularly regarding ionising radiation.

Frequency	Maintenance item	Carried out by
Daily	▪ Cabinet visual check (see page 50)	Customer
	▪ Wafer validation check (see page 56) (optional)	Customer
3 Monthly	▪ Clean and grease the XY manipulator axes	Nordson Test and Inspection
12 Monthly	▪ X-ray leakage check (see page 50)	Customer or Nordson Test and Inspection

The following maintenance items are normally carried out by a Nordson Test and Inspection Customer Service Engineer during an annual machine inspection visit:

- X-ray leakage check
- Machine health check, this includes:
 - Interlock check and test
 - EMO system test
 - HV cable connector check (maintain if necessary)
 - Earth connection check
- Optimize X-ray source (tube)
- Optimize image capture
- Update software and firmware (if beneficial)

7.1. Cabinet visual check

It is recommended that the machine is:

1. Visually checked prior to use to ensure that no damage has occurred to the machine cabinet, and that no unauthorized modifications have been made.
2. Checked during use, for any signs of abnormal operation.
3. Checked during use to ensure the X-ray On indicators at the front, side and rear of the machine are fully functional.

If evidence of damage, or abnormal operation is found:

- Do not use the machine.
- Contact your Nordson Test and Inspection representative for assistance.

7.2. X-ray leakage check

Dage recommend that X-ray leakage is checked every twelve months by using instruments capable of measuring the shielded radiation from a 30-160kV X-ray source to dose rates of less than 1 μ Sv/h.

However, the frequency of X-ray leakage checks can also be determined by reference to any applicable Safe Working Arrangements or Local Rules (see page 69).

The X-ray leakage check procedures described below **must only** be undertaken by suitably-qualified engineers, and/or maintenance technicians who:

- fully understand the safety requirements of the machine, especially with regard to ionizing radiation.
- have received the appropriate training in the use of the required measurement instruments.

An X-ray leakage check can be carried out, as follows:

- a '**Routine Radiation**' dose rate survey - this is carried out at least once a year, as a minimum, or more frequently as determined by the safe working arrangements/local rules.
- a full '**Critical Examination**' dose rate survey - this is carried out under these circumstances:
 - At the installation of the machine.
 - Following **any** modification to the machine that might have damaged or compromised the integrity of the cabinet's lead shielding in any way.
 - If damage to the lead shielding is suspected, either visual damage or known impact.

- After the repair or replacement of any X-ray generating component, any cabinet shielding component or any machine safety system component (for example, an interlock).
- If the machine has been moved/relocated.



X-rays are harmful to the human body.

If radiation leakage in excess of 1µSv/h is detected, switch the machine off and lock the main isolator in the off position immediately.

Seek advice from Dage before using the machine again.

Important considerations when undertaking an X-ray leakage check

- The leakage check procedure is quicker and more thorough if a scintillation type detector is initially used to determine the location of any hot-spots (these are typically indicated by a rapid and localised increase in the displayed CPS). Any detected hot-spots can then be marked and subsequently measured using a dose rate instrument (GM tube).

However, the use of a scintillation type detector is **not** essential.

- When using the dose rate meter **always** allow enough time for the instrument to settle and accurately detect the radiation at each inspection area.
- Checking for radiation leakage from the underside of the cabinet may be deferred depending on installation. If the machine is on the ground floor where there is no physical access underneath, for example no basement or service floor, then it may be sufficient to measure the levels around the 'skirt' of the machine. If the machine is located where it is reasonable to expect that persons below may be exposed to radiation then the levels in this area must be checked. Always consult with your local Radiation Protection Advisor. However, it must be checked at least once per year, or when damage to the shielding is suspected.

Great care should be exercised if the machine has to be raised for this check.

Routine radiation dose rate survey

1. Ensure that all maintenance/service access and electrical compartment doors are properly closed and power-up the machine.
2. Set the wafer chuck centrally in its loading position, and the X-ray tube in the centre.
3. Set the X-ray source to its maximum settings.

Note: If the machine is not regularly used at this voltage then a warm-up may be required.

4. If a scintillation type detector is available, do the following:
 - Carefully scan every part of the cabinet surface with the scintillation detector, paying particular attention to the seams and seals and the overlap between access doors and cabinet.
 - Mark (temporarily, or use a copy of the result sheet (see page 75)) any part of the cabinet where a localised rapid increase in the CPS level is detected and verify with the dose rate meter that the measured radiation level is less than 1 μ Sv/h.
 - If an unusually high CPS reading is detected use the dose rate meter to check the actual level - if safe to do so. If a reading of over 1 μ Sv/h is measured, immediately terminate the survey, switch the machine off and seek corrective action.
5. Using the dose rate meter, do the following:
 - Carefully scan every part of the cabinet surface with the dose rate meter, paying particular attention to the seams and seals, the overlap between access doors and cabinet, and (if applicable) any 'hot-spots' identified by the scintillation detector.
 - Using the result sheet (see page 75) provided, record the location of any measured radiation level higher than 0.2 μ Sv/h.
 - Immediately terminate the survey if a measured radiation reading of over 1 μ Sv/h is detected at any point, switch the machine off and seek corrective action.

Critical Examination dose rate survey

Where necessary, prior to carrying out a critical examination, a dedicated test area around the machine cabinet should be defined (approximately 0.8-2m around the perimeter). Wherever practicable, enclose or identify the test area by physical means, such as; a barrier, warning signs, warning tape, and so on. The only personnel permitted within the defined test area are the personnel actually performing the examination.

1. Ensure that all maintenance/service access and electrical compartment doors are properly closed and power-up the machine.
2. Set the wafer chuck centrally in its loading position, and the X-ray tube in the centre.
3. Set the X-ray source to minimal power (1W), with an initial voltage of 30kV.
4. If a scintillation type detector is available, do the following:
 - Carefully scan every part of the cabinet surface with the scintillation detector, paying particular attention to the seams and seals and the overlap between access doors and cabinet.
 - Mark (temporarily, or use a copy of the result sheet) any part of the cabinet where a localised rapid increase in the CPS level is detected and verify with the dose rate meter that the measured radiation level is less than 1 μ Sv/h.
 - If an unusually high CPS reading is detected use the dose rate meter to check the actual level - if safe to do so. If a reading of over 1 μ Sv/h is measured, immediately terminate the survey, switch the machine off and seek corrective action.
5. Using the dose rate meter, do the following:
 - Carefully scan every part of the cabinet surface with the dose rate meter, paying particular attention to the seams and seals, the overlap between access doors and cabinet, and (if applicable) any 'hot-spots' identified by the scintillation detector.
 - Immediately terminate the survey if a measured radiation reading of over 1 μ Sv/h is detected at any point, switch the machine off and seek corrective action.
6. Increase the X-ray source to maximum power.
7. Repeat steps 4 and 5 above.
8. Increase the tube voltage by 10kV and repeat steps 4 and 5 above.
9. Repeat step 8 up to the maximum tube voltage - 160kV.
10. Using the result sheet (see page 75) provided, record the location of any measured radiation level higher than 0.2 μ Sv/h.
11. On the successful completion of the survey:
 - Complete and sign the record sheet (see page 74) to confirm compliance with the critical examination and store the results in the machine record log.
 - Remove any applicable barriers, tape, or signs.

8. System maintenance operations

8.1. Manually loading a carrier

Wafer carriers (FOUPs) are manually loaded via one of the EFEM's two load ports.



- Load port 1 - left-hand side
- Load port 2 - right-hand side

Note: Depending on system variant, the EFEM load ports may be fitted with access doors. If this is the case, you will need to open the doors prior to loading a FOUP.

1. Place the required FOUP on an available load port.

The EFEM automatically:

- Latches and docks the FOUP.
 - Reads the integral RF tag to extract the Carrier ID for identification purposes.
 - Opens the FOUP and maps the contents.
2. Once the above process has completed, you can verify everything is correct by selecting the Tracking tab (see page 59).

8.2. Logging in

When the machine is first started, the Home page is displayed.

1. If the Home page is not displayed, select the Home tab.
2. If no user is currently logged in, a pop up log in box is displayed.
3. Enter your Engineer-level Username.

This provides the necessary higher level of access to additional functionality, such as recipe creation and machine maintenance.

4. Enter your Password.
5. Select Log In.

6. If an operator-level user is currently logged in, do the following:
 - In the Title Header Bar, select the Logout control.
 - If not already selected, select the Home tab.
 - Log in as an engineer-level user, as detailed above.

8.3. Wafer validation check

This function allows you to manually execute a Wafer Validation check using a designated test wafer in a loaded FOUP carrier. To use this function:

- A test wafer (customer's own, or Nordson Test and Inspection supplied) should be loaded in the same way as production.
- The system must be operating in local mode.
- The required maintenance recipe must have been previously set up.

To manually start the validation check process, do the following:

1. Ensure that the XM8000 cabinet and the EFEM are empty of wafers.
2. Initialise the system by selecting Initialize Components on the Home tab.
3. Wait until the GEM Status pane displays Off-line Local Idle.
4. Place the FOUP loaded with the validation wafer on either of the two load ports.
5. Select Jobs tab > Manual Process Start.

The page displays two identical, but independent load port panes.

6. In the relevant load port pane, select the Maintenance Recipe type.
7. Select ValidationVoidCheck as the required Recipe from the drop down.
8. Set the Use selected slots radio button.
9. Set the required check box in the Slots list.
10. Enter a Lot ID to identify the wafer using the following naming convention:

ValidationVoidCheck_Date

For example: ValidationVoidCheck_18thJAN2019

11. Select Re-use Carrier.

This enables the Start button at the bottom left of the load port pane.

12. Select Start.

The system starts automatically processing the ValidationVoidCheck recipe. The current system State is dynamically displayed.

13. Wait until the validation check run is complete, and the test wafer returns to its original slot.

14. Select Results tab > Results to check the validation run result.

If the void% measurement failed the set tolerance (as agreed with Dage), save the result and send it to the Nordson Test and Inspection XM technical support team using the following email:

XM-DPITechnicalSupport@nordsondage.com.

8.4. Viewing jobs

This function allows you to view both the Process and Control jobs.

A Process Job specifies the process to be applied to a wafer. A Control Job defines what is performed on the contents of a carrier and typically comprises one or more Process Jobs.

1. Select Jobs tab > View.

The Process and Control jobs are displayed in two independent tables.

Each table may extend over more than one page, and certain table columns can be ordered and/or filtered.

2. If you wish to order the table contents, do the following:
3. Select the column heading you wish to sort by (either: Process Job ID or Control Job ID).

The results in the column will be sorted alpha-numerically and the entire table updated accordingly. The column heading shows an order direction icon:

▲ this indicates the order is ascending.

▼ this indicates the order is descending.

4. If you wish to filter the contents of a table column, do the following:

- Select the filter icon in the relevant column heading .
- Select the required filter criteria to apply: Alpha-numeric ascending, alpha-numeric descending, text filter, or text item search.
- Select OK.

8.5. Manually starting a maintenance process

This function allows you to manually execute a maintenance recipe on selected wafers in a loaded carrier. To use this function:

- One or more bare-Si dummy wafer(s) should be loaded in the same way as production.
- The system must be operating in local mode.
- The required recipe must have been previously set up.

To define and manually start a process, do the following:

1. Select Jobs tab > Manual Process Start.

The page displays two identical, but independent Load Port panes.

2. Manually load a carrier on to your chosen load port. The Carrier ID is read and displayed.

Note: The steps described below can be applied to either load port, as required.

3. Select the Maintenance Recipe Type.
4. Select the required Maintenance Recipe from the drop down.

5. If only certain wafers in the loaded carrier are to be included in the job, do the following:
 - Set the Use selected slots radio button.
 - Set the required check boxes in the Slots list. Use the Select/Deselect All controls as necessary.
 - Enter a suitable Lot ID to identify the wafers.
 - Select Apply.
6. If only the occupied slots in the loaded carrier are to be included in the job, do the following:
 - Select the Use occupied slots radio button.
 - Enter a suitable Lot ID to identify the wafers.
7. Select Start.

The system starts automatically processing the wafers. The current system State is dynamically displayed.
8. To abort the manual wafer processing job at any point, click Stop.
9. Select Re-use Carrier to repeat the job.
10. To immediately silence the alarm buzzer, select Stop Buzzer.

8.6. Managing an ad hoc maintenance process

This function allows you to choose which wafers in the loaded carrier are to be included in the job, and the particular maintenance recipe to be used for an individual wafer. Each wafer is then processed a preselected number of times. To use this function:

- One or more bare-Si dummy wafer(s) should be loaded in the same way as production.
- The system must be operating in local mode.
- The required recipes must have been previously set up.

To define and manually execute an ad hoc job, do the following:

1. Select Jobs tab > Ad Hoc.

The page provides two identical, but independent Load Port tabs on the left-hand side, and a process report list on the right-hand side.
2. Manually load a carrier on to your chosen load port. The Carrier ID is read and displayed.

Note The steps described below can be applied to either load port, as required.
3. Select the load port tab you require.
4. If you want to include one or more individually configured wafers in the job, do the following in the slot list:
 - Set the check box against a required wafer entry.
 - Enter the wafer Lot ID.
 - Select the Maintenance Recipe Type.
 - Select the required Maintenance Recipe from the drop down.

- Select the number of Iterations for the wafer.
 - If you want to be able to choose a new recipe after the first recipe run is completed, set the Manual Override check box.
Once the first recipe run is completed (all iterations), a message is displayed prompting you to select a new recipe, or return the wafer to the carrier.
5. If you want to apply a specific configuration to all the wafers, do the following:
 - In the slot list, set the check box against a wafer entry. Typically, this would be entry 1 by default, but can be any entry.
 - Enter the wafer Lot ID.
 - Select the Maintenance Recipe Type.
 - Select the required Maintenance Recipe from the drop down.
 - Select the number of Iterations for the wafer.
 - Select the 'as Slot' from the drop down. This must match the slot entry you have just configured.
 - Select CopyAll.
 6. Select the number of Iterations for the job.
 7. Select Start.
 8. To stop the ad hoc wafer processing job at any point, click Stop.
 9. To abort the ad hoc wafer processing job at any point, click Abort.
 10. Select Re-use Carrier to repeat the job.
 11. To immediately silence the alarm buzzer, select Stop Buzzer.

8.7. Tracking a substrate

This function allows you to track the substrates (wafers) currently loaded in either load port.

Note: Apart from the list selection tabs, there are no user controls on this page. The various displayed fields are for information only and cannot be edited.

1. Select Tracking tab.

The page provides two identical, but independent tabular Load Port Substrate lists on the left-hand side. Each list can be shown or hidden, as required. However, you can only show one list at a time.

A Current Substrate report is displayed on the right-hand side, it shows a number of substrate parameters, job statistics and equipment status information.

2. Select the required Load Port Substrates tab.

The tabular list shows the Location, State and Jobs for each substrate (wafer).

The substrate can be in one of four Locations:

- A numbered carrier slot.
- Tool - this is the X-ray machine wafer chuck.
- Robot - this is the EFEM End Effector robotic wafer handler.
- Aligner - this is the EFEM aligner mechanism.

8.8. Maintenance control tab

The maintenance tab allows an engineer-level user to enter and exit the maintenance mode. Simply select the Enter or Exit control, as required.

In addition, this tab also provides access to the maintenance-based functions listed in the table below.

Note: Access to the maintenance tab is only allowed when a job is not running, and the X-rays are off.

Function	This provides access to:
Equipment Status	the following dynamically-updated X-ray parameters: ▪ Voltage (kV) ▪ Power (W) ▪ Uptime (hours)
Manual Control	dynamically-updated machine control status information, and the controls necessary for allowing the manual operation of various system components, such as the EFEM: ▪ load ports ▪ robot ▪ aligner ▪ shutter
Recovery	the controls that allow you to stop and initialize the five major sub-systems classified as machine 'components'.
Door Control	the controls that allow you to open and close the X-ray enclosure maintenance/service access doors.
IO Control	a range of function-specific machine indicators and manual controls for the: ▪ Shutter ▪ Wafer Chuck ▪ Door
Load Port Usage	manual controls and state indicators for both load ports.
Equipment Constants	a number of machine-dependant variables that can be altered to support different operational situations.

8.8.1. Equipment status

The equipment status page provides dynamically-updated machine information, as detailed below:

- X-Ray Voltage (kV)
- X-Ray Power (W)
- X-Ray Uptime (hours)

8.8.2. Manual control

The manual control page is divided into five discrete sections. Each section is dedicated to a particular machine functional area and provides both dynamically-updated status information, and the controls necessary for allowing manual operation of the various system components and robotics, as detailed in the tables below.

Note: Each section has its own message pane that displays any applicable system messages.

When used together, the controls provided allow you to carry out a range of tasks such as:

- Clamping, loading and mapping a carrier.
- Moving an individual wafer from a source location to a chosen destination location.
- Opening and closing the EFEM to X-ray cabinet shutter, and loading or unloading the wafer chuck.

1 - Load Port 1 + 2 Control section

Display/Control	Description
Re-use Carrier	Select the appropriate control to repeat the maintenance task using the currently loaded carrier.
Clamp/Unclamp Carrier	Select the appropriate control to clamp or unclamp the carrier (FOUP) currently loaded. The current clamp status is displayed: ▪ Unclamped ▪ Clamped
Load/Unload Carrier	Select the appropriate control to load or unload the carrier (FOUP) currently loaded. The carrier is docked, opened and mapped ready for use. The current load status is displayed: ▪ NO CARRIER ▪ Loaded

Display/Control	Description
Map Status	This displays the current carrier map status: - Unknown - Mapped

2 - Robot Control section

Display/Control	Description
Source:	Select the source location of the wafer you wish to use. The prefixes LP1 and LP2 are used to identify specific load ports. The list includes only system locations where a wafer is actually present, such as: - Mapped load port carrier slot - Aligner - Inspection (X-ray machine wafer chuck) The list also includes the substrate (wafer) ID.
Destination:	Select the destination location of the wafer you have selected. The list includes only system locations where no wafers are present, such as: - Empty load port carrier slot - Aligner - Inspection (X-ray machine wafer chuck) Notes: - It is recommended that wafers are ALWAYS pre-aligned prior to being passed to the X-ray machine. - If the shutter is closed, the Inspection location is not available.
Move	Select to move the wafer from the source to the destination location. Once a move operation has completed, the source, destination and occupancy display fields on this page are dynamically updated.

3 - PreAligner section

Display/Control	Description
Occupancy:	This displays the current occupancy status of the aligner: - EMPTY - Current substrate (wafer) ID
Value field	Select the alignment angle value

Display/Control	Description
AlignWafer	Select to align the wafer. The current alignment status is displayed: - Unknown - Aligned

4 - Inspection Control section

Display/Control	Description
Occupancy:	This displays the current occupancy status of the X-ray machine: - EMPTY - Current substrate (wafer) ID
Load/Unload	Select to load/unload the wafer chuck.
Inspection	Moves the wafer chuck to the inspection position (directly under the X-ray tube).

5 - Shutter Control section

The shutter can only be operated manually when the machine's safety system requirements have been fully met.

Display/Control	Description
Open Shutter	Select to open the EFEM to X-ray cabinet shutter.
Close Shutter	Select to close the EFEM to X-ray cabinet shutter.

8.8.3. Recovery

The recovery page allows you to stop and initialize the five major sub-systems classified as machine 'components', using the manual controls detailed in the table below.

Note: The initialize and stop functions provided here are the same functions as provided on the Home tab.

Control	Description
System Recovered	Select to attempt to recover the system from a failure state. However, in most cases it is recommended that you stop the components and re-initialize when a failure occurs (see below).
Initialize Components	Select to initialize the five machine components. If a component fails to initialize, the problem needs to be investigated before you can continue. Once a component has successfully initialized, the relevant entry in the on-screen component list display is updated from Stopped to Running.

Control	Description
Stop Components	Select to stop the five machine components. This function terminates all machine operation. However, this action does not immediately cease machine operation or mechanical movement, as the current activity of each component is first completed, before it becoming inactive.
Shutdown PCs	Select to shutdown the three system PCs (WFC, IO, INS).

8.8.4. Door control

The door control page provides the manual controls that allow you to open (unlock) and close (lock) the X-ray enclosure maintenance/service access doors.

Control	Description
Open Door	Select to unlock the maintenance/service access doors. This function simply unlocks the door interlock. The two bolts fitted to the right-hand door will still need to be removed, and the doors physically opened. All X-ray generation is terminated, and the machine put into a non-recoverable state that requires a complete restart.
Close Door	Select to lock the maintenance/service access doors. This function simply locks the door interlock. The doors will still need to be physically closed and the right-hand door secured by the front-mounted bolts.

8.8.5. IO Control

The IO Control page provides a number of function-specific machine IO state indicators, and the manual controls detailed in the tables below.

Note: The shutter and door can only be operated manually when the machine's safety system requirements have been fully met.

The IO indicators allow you to see any changes in state dynamically during the manual exercising of the shutter, wafer chuck and door locks, as well as providing useful diagnostic machine state information in the event of a failure.

Shutter and wafer chuck

Control	Description
Open	Select to open the shutter, or wafer chuck.
Close	Select to close the shutter, or wafer chuck.
Cycle	Select to start a cycle of open and close operations.
Stop	Select to stop any currently running cycle operation.

Door

Control	Description
Open Door	Select to unlock the maintenance/service access doors. This function simply unlocks the door interlock. The two bolts fitted to the right-hand door will still need to be removed, and the doors physically opened.
Close Door	Select to lock the maintenance/service access doors. This function simply locks the door interlock. The doors will still need to be physically closed and the right-hand door secured by the front-mounted bolts.

8.8.6. LOAD port usage

The load port usage page provides the following manual controls and indicators for both load ports:

Control/Indicators	Description
In Service	Select to set the load port as available for use.
Out of Service	Select to set the load port as unavailable for use. You would typically use this to indicate if there is a fault with the load port.
Access Mode: Auto	Select to enable the automatic transfer of the carriers (FOUPs) to and from the load port via an Automatic Material Handling System (AMHS).
Access Mode: Manual	Select to enable carriers to be manually placed and removed from the load port by a local user.
Continue Manual Carrier Delivery	Select to allow a carrier that has been manually placed on a load port currently in Auto access mode, and with a transfer state Ready To Load - to be processed. Normally, when this situation arises, the system raises an access violation alarm and treats the transfer as incomplete. However, this function allows you to proceed with manual operation.
PIO Reset	Select to clear any PIO alarms related to the load port.
Hardware PIO	This shows the state of the various inputs and output lines used by the load port and the AMHS to control carrier transfer between them.

8.8.7. Equipment constants

If enabled, the host can read and update a number of equipment constant. This page provides information on, and allows the enabling of these specific equipment constants parameters.

Note: These parameters must not be changed unless under the strict guidance of Nordson Test and Inspection.

Control	Description
Drop down	This lists the equipment constants, select as required. Set the value to: ▪ 0 = Disabled ▪ 1 = Enabled
Update	Select to save the change.

9. Troubleshooting

Every customer facility will have its own methods for problem recovery. Here we provide basic troubleshooting methodology.

The table below is by no means exhaustive, however its purpose is to provide you with a logical starting point.

When a problem occurs:

- Please refer to the most relevant symptom below in order to progress to the next stage of troubleshooting.
- If there are several alarms and/or errors at the same time, it may be difficult to identify the one which is the root cause of the problem.
- In all cases, aim to identify the root cause by a process of verification and elimination.

No.	Symptom	Possible Problem Area
1	System fails to power ON	Verify power from facilities. Verify that all EMO buttons are out. Verify that the key switch on the main control panel is in the Power Enable or X-ray Enable position. Verify that the green power on button illuminates. Verify all door interlocks are made (doors are shut correctly). If still no power, contact Nordson Test and Inspection service engineer.
2	System powers OFF	Try powering system back on and if successful, verify alarm logs. If unsuccessful, follow item 1 above.
3	System reports vacuum error	Verify vacuum supply from facilities. Check Pneumatic settings.
4	System reports low air pressure	Verify compressed air (CDA) supply from facilities. Check Pneumatic settings.
5	Wafer slipping from robotics handling	Recover wafer(s) manually using Manual Controls (see page 61). Check items 3 and 4 above. Consider running bare-Si dummy wafers to verify using Manual Controls.

No.	Symptom	Possible Problem Area
6	Timeout alarms for wafer handling robotics	Recover wafer(s) manually using Manual Controls (see page 61). Check items 3 and 4 above. Consider running robotics manually using Manual Controls.
7	Timeout alarm for shutter open / close	Recover wafer(s) manually using Manual Controls (see page 61). Check item 4 above. Consider running shutter manually using Manual Controls.
8	Wafer analysis stops unexpectedly	STOP and take a note of the alarm and/or error message. Recover wafer(s) back to FOUP(s) using Manual Controls (see page 61). Using alarm and/or error information, check the most relevant item(s) above. Consider performing an early Clean and grease axes.
9	Imaging / X-ray on, or analysis problems	Verify that the key switch on the main control panel is in the X-ray Enable position. Consider performing an early X-ray source exchange.
10	Any other errors generated by the system during normal operation	STOP and take a note of the alarm and/or error message. Contact Nordson Test and Inspection service engineer.

If the problem is not covered in the table above, then it will require more detailed troubleshooting, involving checking individual circuits and systems.

As a result it is strongly advised that you contact your local Nordson Test and Inspection office for additional support.

For any issues arising specifically related to the EFEM, please consult the specific EFEM user manual for additional troubleshooting help.

10. Appendix 1 - Radiation Safety

10.1. Safe working arrangements

In all matters regarding radiation safety you must always take due account of any radiation protection legislation in your own country, and comply with all applicable local and company regulations.

If you are unsure of any of these, **always** seek professional advice.

VERY IMPORTANT considerations:

- The safe working arrangement instructions and procedures referred to in the numbered sections below are based on UK regulations in force at the time of printing (which may or may not be appropriate in other countries). They have been provided **SOLELY FOR GUIDANCE ONLY** and are expected to be modified to suit your local requirements as they simply provide examples of the type of content you may choose to include in your instructions, rules and procedures.
- It is the **SOLE** responsibility of the customer to produce any safe working arrangement instructions, rules or procedures required for the operation and use of the X-ray machine.
- The inclusion of any example content provided here, or any new content, will depend on national legislation, local and company regulations.
- Always seek local professional advice when preparing your instructions, rules and procedures.

1 - Locale and machine type

The following could be included here:

- The details of the applicable X-ray machine, or machines.
- The address of the establishment where the applicable X-ray machine, or machines are installed.

2 - Personnel authorised to use the equipment

The following could be included here:

- A statement that only authorised personnel who have received suitable training may use this equipment.
- A list naming all the authorised personnel.

3 - Appointed safety personnel

Depending on national legislation, local and company regulations, the appointment of the following may be a requirement:

- One, or more on-site **Radiation Protection Supervisors (RPS)**.
- One, or more external **Radiation Protection Advisors (RPA)**, or **Qualified Radiation Experts (QRE)**.

Typically, an RPA/QRE advises the company in the observance of the national and local regulations, and in any health and safety matters in connection with ionising radiation.

Please note, depending on nationality, the titles stated above could be different.

Where appointed, the names and applicable contact details (including full address) of the above personnel could be listed here (including any back-up contacts).

4 - Warnings

A warning similar to this could be included here:

- X-rays can be harmful to human health. If a fault condition arises or the integrity of the shielding of the X-ray machine is suspect, then switch off immediately and seek further advice from your appointed Radiation Protection Supervisor (RPS).

5 - Cautionary statements

Statements similar to the following examples could be included here:

- Everyone is reminded of their duty of co-operation, consideration and communication with other employees or persons as stated in the company's Health and Safety Regulations (or similar, as applicable).
- Failure to observe these safe working arrangements will be reported to the company's management team, which may lead to disciplinary action, or summary dismissal.
- If you are unsure about any procedure detailed in the safe working arrangements, contact an RPS before commencing work with the equipment.
- Do not use equipment that is known to be faulty, or where there is any doubt about its safe operation.

6 - Additional contacts

Where appointed, the names and applicable contact details of the following (or similar, as applicable) could be listed here:

- Company Health and Safety personnel.
- Company Fire Officer.
- Company Medical Officer and/or Advisor.
- Out-of-hours emergency contact.

7 - Fire instructions

The company's instructions for the X-ray machine operators in the event of a fire could be listed here, for example:

- In the event of fire, operate the Emergency Stop pushbutton (provided there is no additional personal risk to you or others) and follow the company's established fire evacuation drill.

8 - Lock-out and key control instructions

The company's instructions regarding the safe isolation of the machine for servicing could be listed here. Please refer to the Lock Out - Tag Out section of this guide for full information (see page 39).

9 - Machine record log

It is advisable to record all actions and procedures carried out on the X-ray machine using a suitable 'record log'. This should stay with the applicable machine and be available to all personnel operating or servicing the machine.

10 - Safety check schedule

The following inspections and safety checks are recommended (as a minimum):

- Daily - It is recommended that the machine is visually checked for damage prior to use to ensure integrity of the cabinet shielding and maintenance/service access doors. During use, check for any signs of abnormal operation. If evidence of damage, or abnormal operation is found:
 - Do not use the machine.
 - Contact your Nordson Test and Inspection representative for assistance.
- Daily - During use observe that the X-ray On indicators are fully functional.
- Annually - Undertake a routine radiation dose rate survey.

However, the frequency of these checks may be subject to variation depending upon local regulations.

Please refer to the Routine preventative maintenance section of this guide for full information (see page 49).

The following statements and warning could be included here, for example:

- Consult an RPS if a problem is identified, or suspected during:
 - Normal operation.
 - Any scheduled safety checks undertaken on a working X-ray machine.
- Do not use equipment that is known to be faulty, or where there is any doubt about its safe operation. **Always** seek corrective action and/or contact your Nordson Test and Inspection representative for assistance.
- If a unit is not in use, this should be recorded on the machine record log.

11 - Test limitations

As a minimum, the following statement could be included here:

- The X-ray machine must only be used with a maximum X-ray tube voltage of 160kV (Max target power for system). Any use outside these values is **prohibited**.

12 - Normal operation and maintenance

As a minimum, the following statement could be included here:

- Refer to the User and Maintenance Guides.

13 - Modifications

A statement or instructions regarding making modifications to the X-ray machine could be included here, for example:

- The enclosure and associated components should not be modified or added to in any way without the express permission and approval of Nordson Test and Inspection. Documentation of all modifications, once approved and followed by a Critical Examination, must be filed in the machine record log.

14 - Accidental exposure

The following statements could be included here:

- If you suspect that yourself or others have been inadvertently exposed to X-rays as a result of an interlock failure or significant leakage through the cabinet shielding (for example) do the following:
 - Switch off and isolate the equipment (if not already off).
Take care when isolating the equipment not to increase your exposure.
 - Report the incident to the RPS immediately.
- If a suspected accident or other occurrence is reported then the RPS must undertake an immediate investigation.

15 - Dose investigation level

Depending on national legislation, local and company regulations, the following statements (or similar) could be included here - wherever applicable:

- The ionising radiation regulations (give the applicable regulation references here) require the company to carry out an investigation when an employee receives an effective dose for the first time in the calendar year of (give the applicable level here) or such lower effective dose as the company may specify with the assistance of the RPA/QRE. The purpose of this investigation is to determine whether doses are being kept as low as reasonably practicable.
- The company will carry out an investigation when an employee receives an effective dose greater than the set limit, this dose is called the investigation level. This investigation will attempt to establish the reason for the received dose in order to determine whether further controls are required to restrict exposures. Any investigation shall be undertaken with the assistance of the RPA/QRE.

16 - Over exposure

Depending on national legislation, local and company regulations, the following statements (or similar) could be included here - wherever applicable:

- If the investigation shows beyond any reasonable doubt that an overexposure has occurred (greater than the set limit) then additionally the following should be informed, and a detailed investigation begun:
 - The local Health and Safety Executive (HSE, or in-country equivalent).
 - The company's Medical Advisor.
- Copies of the investigation should be retained for (give the applicable period here) years or until the employee is (give the applicable age here), if longer. The person(s) involved should be informed of the results of any investigation.

10.2. Critical examination record sheet

The following tables and diagrams can be enlarged and used as critical examination and radiation level record sheets:



XM8000
Intelligent X-ray Metrology

Critical Examination Certificate

☐
Tick appropriate box

Routine Radiation Survey

☐

Machine Type	XM8000	Machine Serial No	
Test Date		Measured By (Print Name)	

	Passed	Name	Date
Emergency Stop Check disables X-rays	☐	Passed By:	
Interlocks Check prevents X-ray operation	☐	Passed By:	
Warning Lights Check all lights operate on/off	☐	Passed By:	
Visual HV Cable Check Undamaged and correct gap set	☐	Passed By:	
Warning Signs/Labels Check presence	☐	Passed By:	
High Voltage PSU Check correct unit is fitted	☐	Passed By:	

HVPSU Model Number	
Tube Type	NT MK3
Tube Voltage (Max.)	160 kV

HVPSU Serial Number:	
Tube Serial Number:	
Tube Power (Max)	W

Radiation Meters	Model	T Number (if available)	Serial Number	Calibration Expiry Date
Radiation Scintillation Meter				
Radiation Dose Meter				

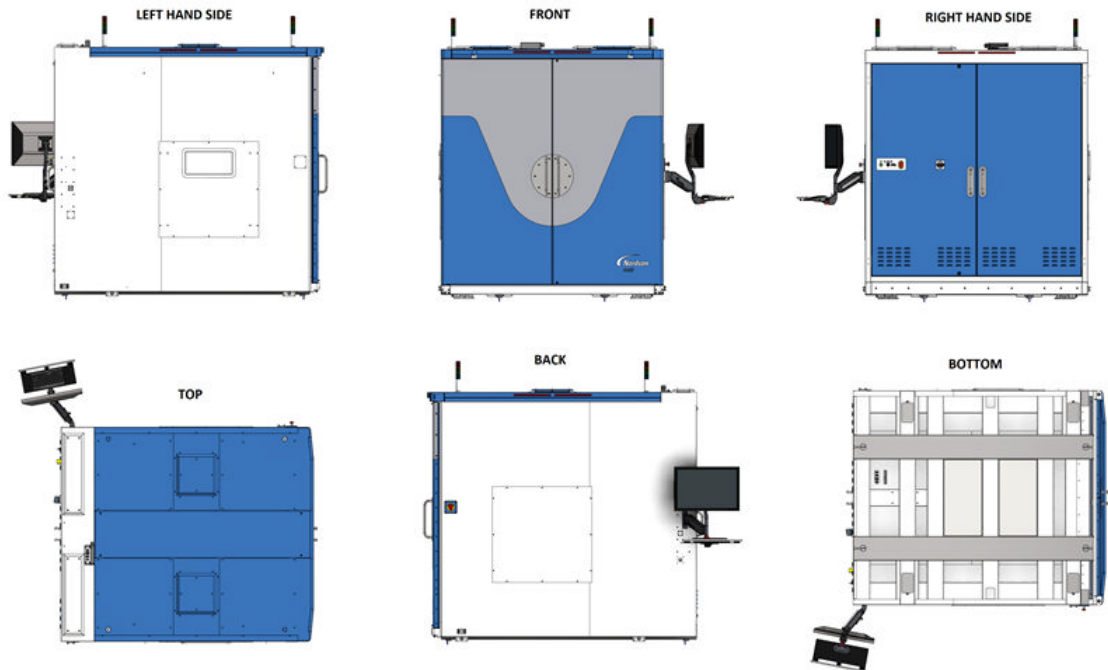
Maximum Measurement		μSv/h	Background Radiation		μSv/h
----------------------------	--	-------	-----------------------------	--	-------

Measured levels (Units - $\mu\text{Sv/h}$)

Set the wafer chuck centrally in its loading position, and the X-ray tube in the centre. Scan the entire outside surface of the cabinet. After reaching 160kV and (max) W, record all readings exceeding $0.2\mu\text{Sv/h}$ on the diagram below using numbered circles.

Record the X-ray intensity using a GM tube dose rate meter for all identified points.

Stop immediately and investigate any readings above $1\mu\text{Sv/h}$.



All readings are below $1\mu\text{Sv/h}$

Passed By:

Date:

Measure X-ray intensity on the cabinet across the whole surface for initial test shown above.
Use the dose meter and log measurements exceeding **$0.2\mu\text{Sv/h}$** in the table below. Units: $\mu\text{Sv/h}$

1		4		7		10		13		16	
2		5		8		11		14		17	
3		6		9		12		15		18	

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